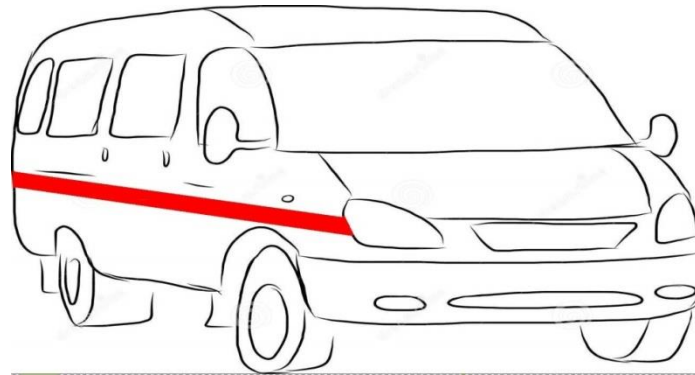




First Digital Tunisian Louage's Transportation Solution



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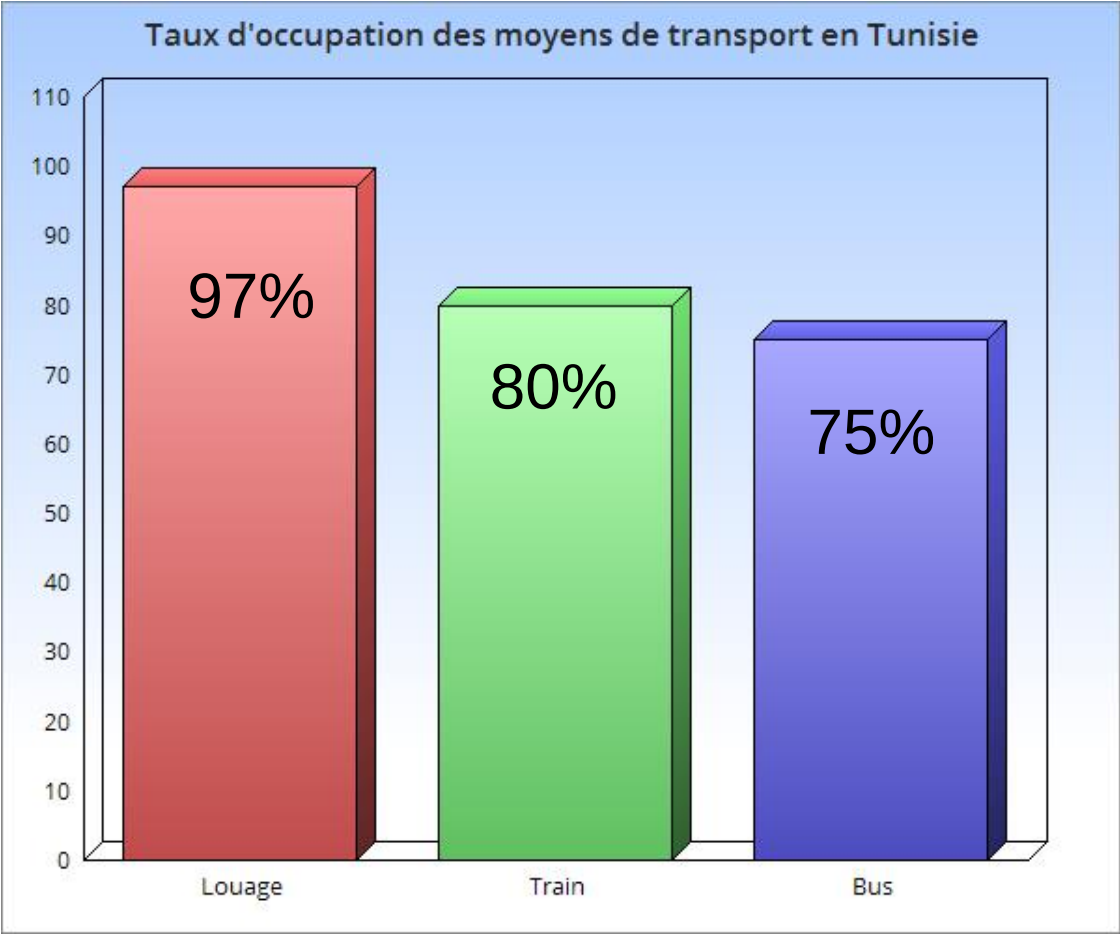
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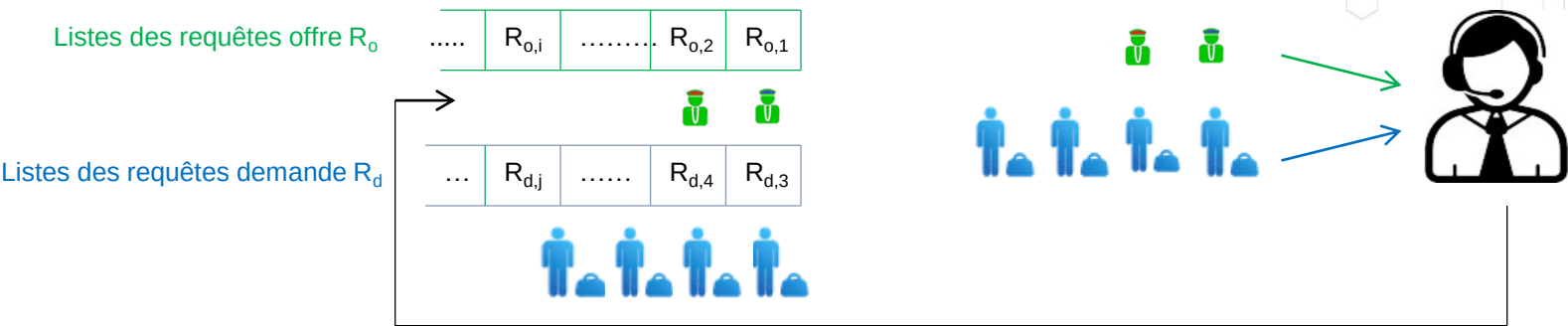
1. LOUAGE'S SYSTEM IN TUNISIA
2. THE 1ST FORMALIZATION OF LOUAGE
 1. STATE OF THE ART
 - 1.1 ANALOGOUS SYSTEMS
 - 1.2 AN INCREMENTAL EVOLUTION OF DARP, VRP, TSP
 2. DEMAND & OFFER REQUESTS, ORGANIZATIONS
 3. A PRAGMATIC EXAMPLE OF PAIRING REQUESTS
 4. MULTI CONSTRAINTS
 5. MULTI OBJECTIVES
3. PROPOSED ALGORITHM AND TESTS
 1. GILA : GREEDY INCREMENTAL LOUAGE'S ALGORITHM
 2. PRAGMATIC EXAMPLE OF PAIRING REQUESTS APPLYING GILA
 3. TESTS AND RESULTS ANALYSIS
4. CONCLUSIONS AND PERSPECTIVES



- 10000 vehicles of louage in 2014.
- 15 % of km/ individual vehicles made by louage.
- Very cheaper transportation : 4.75 dinars / 60 km (i.e. 0,04\$ per km).
- And an outstanding occupation rate, see below !



Conclusions and perspectives



- ✚ A **driver**, arriving at the station, register his line to be operate to an operator (if present; otherwise it's auto-régulation).
- ✚ Passengers register their request demands.
- ✚ The operator has to pair offers and demands respectfull to rules and pragmatism.

Individuel taxi



Collective taxi



Uber



Rural Transport



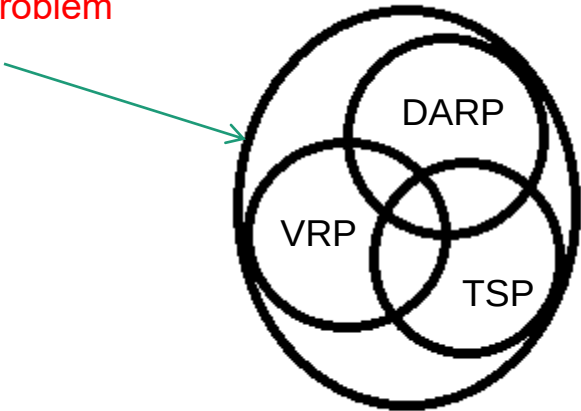
VTC



+ Louage's problem is defined by multi constraints and multi objectives.

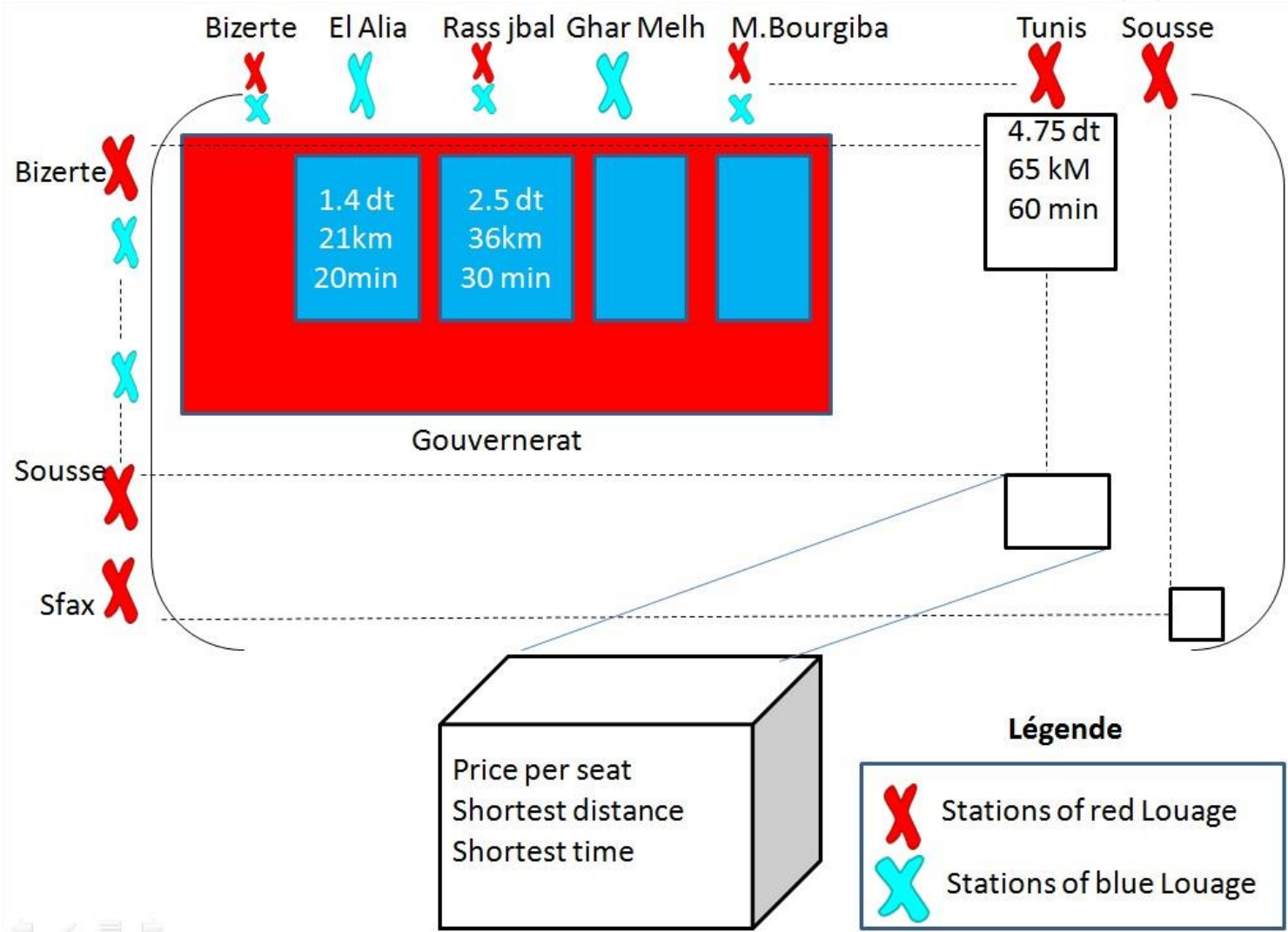
Louage's problem

+ We distinguish several classical combinatorial problems applied to Transportation domain :
TSP: traveling salesman problem
VRP: vehicle routing problem
DARP: Dial a ride problem



Conclusions and perspectives

Matrix origin-destination



✚ Passengers send a list of demand requests (Demands).

$$Demands = \begin{cases} R_{d,1} \\ \vdots \\ R_{d,j} = Itin_{d,j}, NP_{d,j}, M_{d,j}, State_{d,j} \\ \vdots \\ R_{d,\overline{R_d}} \end{cases}$$

✚ $Itin_{d,j}$ designates the initial path or itinerary.

$$Itin_{d,j} = \{Pos_{d,j,1}[h_{d,1}^-, h_{d,1}^+, PU_{d,1}, D_{d,1}], \dots, \\ Pos_{d,j,\overline{Itin_{d,j}}}[h_{d,\overline{Itin_{d,j}}}^-, h_{d,\overline{Itin_{d,j}}}^+, PU_{d,\overline{Itin_{d,j}}}, D_{d,\overline{Itin_{d,j}}}] \}$$

✚ $NP_{d,j}$ number of booked places

✚ $M_{d,j}$ the bill.

✚ $State_{d,j}$ designates the state of the request : initialized, validated, paired, contractualized, running, realized, non realized and archived).

✚ Driver (or Chauffeur in this slide) defines the set of all driver

$$\text{Chauffeur} = \{\text{Chauffeur}_c \text{ telque } c \in [1, \overline{C}]\} \text{ avec}$$
$$\text{Chauffeur}_c = (\text{matricule}, \text{type de permis}, \text{lignes}_c \dots)$$

✚ Idem for Vehicle as a set of vehicles

$$\text{Véhicule} = \{\text{Véhicule}_v \text{ telque } v \in [1, \overline{V}]\} \text{ avec}$$
$$\text{Véhicule}_v = (\text{capacité}, \text{assuré}, \text{consommation}, \text{conducteurs} \dots).$$

✚ Drivers interact with the system by declaring their Offers.

$$Offers = \begin{cases} R_{o,1} \\ \vdots \\ R_{o,i} = Itin_{o,i}, C_{o,i}, V_{o,i}, PlacesL_{o,i,initial}, PlacesL_{o,i,actuel}, State_{o,i} \\ \vdots \\ R_{o,\overline{R_o}} \end{cases}$$

✚ Each Offer request is composed of :

-driver's itinerary :

$$Itin_{o,i} = \{Pos_{o,i,1}[h_{o,1}^-, h_{o,1}^+, PU_{o,1}, D_{o,1}], \dots, \\ Pos_{o,i,\overline{Itin_{o,i}}}[h_{o,\overline{Itin_{o,i}}}^-, h_{o,\overline{Itin_{o,i}}}^+, PU_{o,\overline{Itin_{o,i}}}, D_{o,\overline{Itin_{o,i}}}] \}$$

- $C_{o,i}$: driver associated to offer $R_{o,i}$.

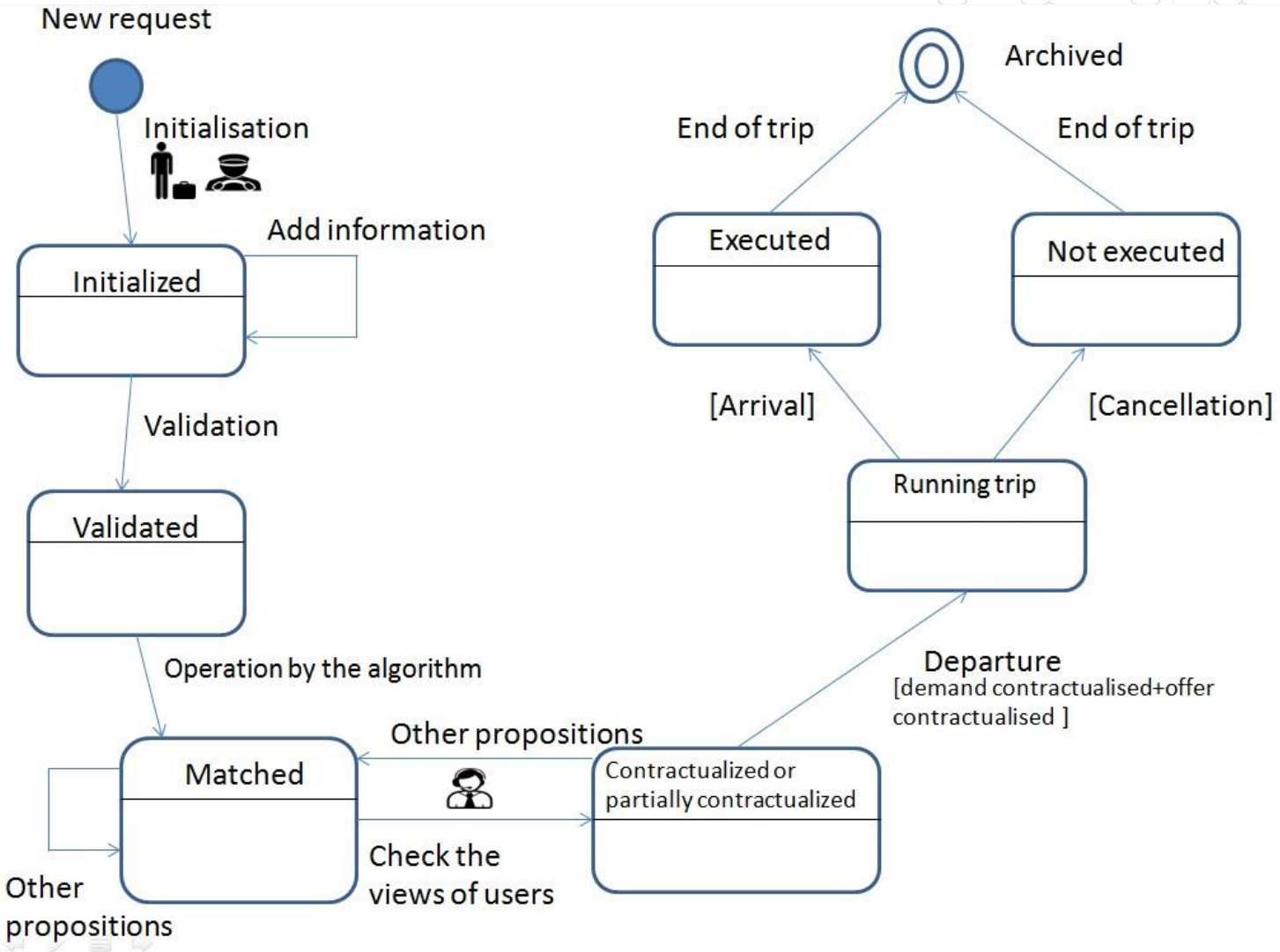
- $V_{o,i}$: vehicle v associated to request offer $R_{o,i}$

- $PlacesL_{o,i,initial}$: number of available seats at initialization state of $R_{o,i}$.

- $PlacesL_{o,i,actuel}$: number of actual available seats at current time (and then state) of $R_{o,i}$.

- $State_{o,i}$: status of the request.

State-Transition Diagram of requests



Organizations are obtained consecutively of pairing offer(s) and their demand.

$$Orgs = \begin{cases} Org_1 \\ \vdots \\ Org_{Cse} = \begin{cases} Cse_1 \\ \vdots \\ Cse_k = Itin_{Cse,k}, Ro_{Cse,k}, ListRd_{Cse,k}, State_{Cse,k}, PlaceO_{Cse,k}, PlaceL_{Cse,k} \\ \vdots \\ Cse_{\overline{Org_{Cse}}} \end{cases} \\ \vdots \\ Org_{\overline{Orgs}} \end{cases}$$

- $Itin_{Cse,k}$, itinerary of journey k

$$Itin_{Cse,k} = \{Pos_{Cse,k,1}, \dots, Pos_{Cse,k,p}, \dots, Pos_{Cse,k, \overline{Itin_{Cse,k}}}\}$$

- $Ro_{Cse,k}$, offer paired with journey k.

- $ListRd_{Cse,k}$: liste of demands paired with an offer.

$$ListRd_{Cse,k} = \{Rd_{Cse,k,1}, \dots, Rd_{Cse,k,l}, \dots, Rd_{Cse,k, \overline{ListRd_{Cse,k}}}\}$$

- $State_{Cse,k}$: indicates the status of journey k.

- $PlaceO_{Cse,k}$: indicates the maximal number of seats occupied by passengers at any step of the itinerary of Cse_k .

- $PlaceL_{Cse,k}$: indicates the minimal number of seats occupied by passengers at any step of the itinerary of Cse_k .

Requests pairing example

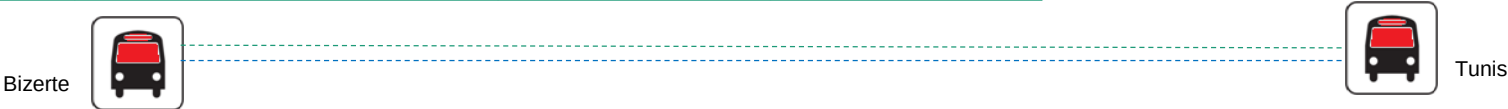
$R_{d,8}$ $\begin{pmatrix} 6:20 & 7:20 \\ 6:20 & 7:20 \end{pmatrix}$ Bizerte	$\xrightarrow{+4}$	$\xrightarrow{-4}$ $\begin{pmatrix} 7:20 & 8:20 \\ 7:20 & 8:20 \end{pmatrix}$ Tunis
$\xrightarrow{+2}$ $\begin{pmatrix} 6:20 & 7:20 \\ 6:20 & 7:20 \end{pmatrix}$ Bizerte	$\xrightarrow{-2}$ $\begin{pmatrix} 7:45 & 8:30 \\ 7:45 & 8:30 \end{pmatrix}$ Tunis	$R_{d,7}$
$\xrightarrow{+30}$ $\begin{pmatrix} 6:45 & 7:30 \\ 6:45 & 7:30 \end{pmatrix}$ Bizerte	$\xrightarrow{-30}$ $\begin{pmatrix} 7:15 & 8:15 \\ 7:15 & 8:15 \end{pmatrix}$ El Alia	$R_{d,6}$
$\xrightarrow{+3}$ $\begin{pmatrix} 6:15 & 7:15 \\ 6:15 & 7:15 \end{pmatrix}$ Bizerte	$\xrightarrow{-3}$ $\begin{pmatrix} 7:45 & 8:30 \\ 7:45 & 8:30 \end{pmatrix}$ Tunis	$R_{d,3}$
$\xrightarrow{+4}$ $\begin{pmatrix} 6:30 & 7:45 \\ 6:30 & 7:45 \end{pmatrix}$ Bizerte	$\xrightarrow{-4}$ $\begin{pmatrix} 6:30 & 8:45 \\ 6:30 & 8:45 \end{pmatrix}$ Tunis	$R_{d,2}$

Request demand



$R_{o,5}$ $\begin{pmatrix} 6:30 & 7:30 \\ 6:30 & 7:30 \end{pmatrix}$ Bizerte	$\xrightarrow{+1}$	$\xrightarrow{0}$ $\begin{pmatrix} 6:45 & 7:45 \\ 6:45 & 7:45 \end{pmatrix}$ El Alia	$\xrightarrow{-1}$ $\begin{pmatrix} 7:00 & 8:00 \\ 7:00 & 8:00 \end{pmatrix}$ Rass Jbel	$R_{o,4}$ $\begin{pmatrix} 6:45 & 7:30 \\ 6:45 & 7:30 \end{pmatrix}$ Bizerte	$\xrightarrow{+3}$	$\xrightarrow{-2}$ $\begin{pmatrix} 7:00 & 8:15 \\ 7:00 & 8:15 \end{pmatrix}$ Rass Jbel	$R_{o,1}$ $\begin{pmatrix} 6:45 & 7:15 \\ 6:45 & 7:15 \end{pmatrix}$ Bizerte	$\xrightarrow{+1}$	$\xrightarrow{-1}$ $\begin{pmatrix} 6:45 & 8:15 \\ 6:45 & 8:15 \end{pmatrix}$ Tunis
$PI=30$				$PI=4$			$PI=8$		

Request offer



Legend

Trajet de passager

Trajet de chauffeur

Passager

Chauffeur d'un louage bleu

L'opérateur

Chauffeur d'un louage rouge

Louage à bande bleu

Louage à bande rouge

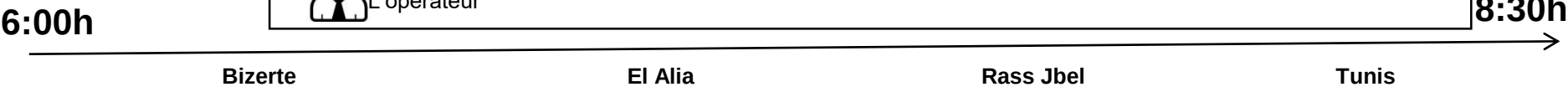
R_d Requête demande

R_o Requête offre

PI Places libres

$+1$ Un chauffeur ou un passager montant (PU)

-1 Un chauffeur ou un passager descendant (D)



Requests pairing example

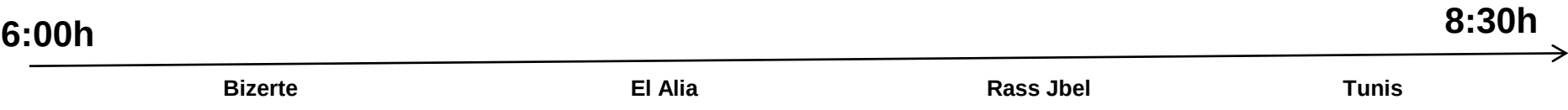
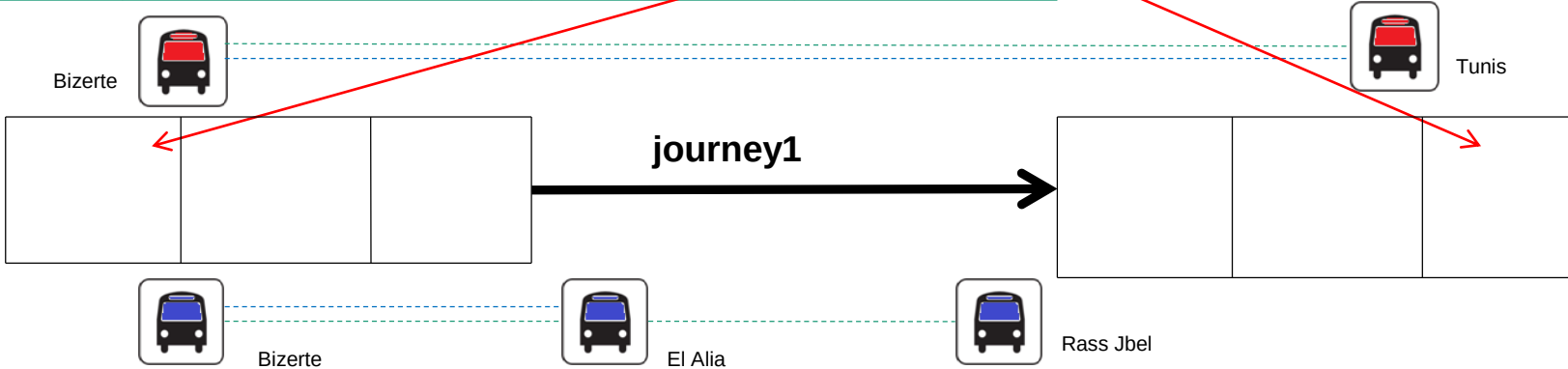
$R_{d,8}$ $\begin{pmatrix} +4 \\ 6:20 & 7:20 \end{pmatrix}$ Bizerte $\rightarrow$ $\begin{pmatrix} -4 \\ 7:20 & 8:20 \end{pmatrix}$ Tunis	$R_{d,7}$ $\begin{pmatrix} +2 \\ 6:20 & 7:20 \end{pmatrix}$ Bizerte $\rightarrow$ $\begin{pmatrix} -2 \\ 7:45 & 8:30 \end{pmatrix}$ Tunis	$R_{d,6}$ $\begin{pmatrix} +30 \\ 6:45 & 7:30 \end{pmatrix}$ Bizerte $\rightarrow$ $\begin{pmatrix} -30 \\ 7:15 & 8:15 \end{pmatrix}$ El Alia	$R_{d,3}$ $\begin{pmatrix} +3 \\ 6:15 & 7:15 \end{pmatrix}$ Bizerte $\rightarrow$ $\begin{pmatrix} -3 \\ 7:45 & 8:30 \end{pmatrix}$ Tunis	$R_{d,2}$ $\begin{pmatrix} +4 \\ 6:30 & 7:45 \end{pmatrix}$ Bizerte $\rightarrow$ $\begin{pmatrix} -4 \\ 6:30 & 8:45 \end{pmatrix}$ Tunis
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Request demand

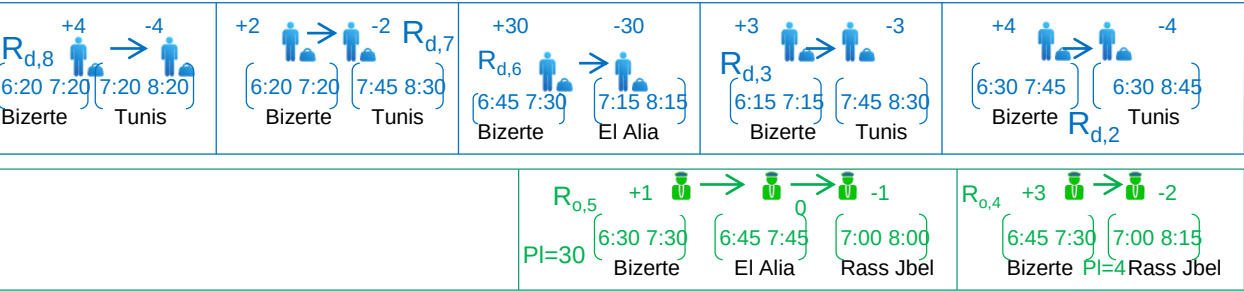


$R_{o,5}$ $\begin{pmatrix} +1 \\ 6:30 & 7:30 \end{pmatrix}$ Bizerte $\rightarrow$ $\begin{pmatrix} 0 \\ 6:45 & 7:45 \end{pmatrix}$ El Alia $\rightarrow$ $\begin{pmatrix} -1 \\ 7:00 & 8:00 \end{pmatrix}$ Rass Jbel $PI=30$	$R_{o,4}$ $\begin{pmatrix} +3 \\ 6:45 & 7:30 \end{pmatrix}$ Bizerte $\rightarrow$ $\begin{pmatrix} -2 \\ 7:00 & 8:15 \end{pmatrix}$ Rass Jbel $PI=4$	$R_{o,1}$ $\begin{pmatrix} +1 \\ 6:45 & 7:15 \end{pmatrix}$ Bizerte $\rightarrow$ $\begin{pmatrix} -1 \\ 6:45 & 8:15 \end{pmatrix}$ Tunis $PI=8$
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Request offer



Requests pairing example



Request demand



Request offer



Requests pairing example

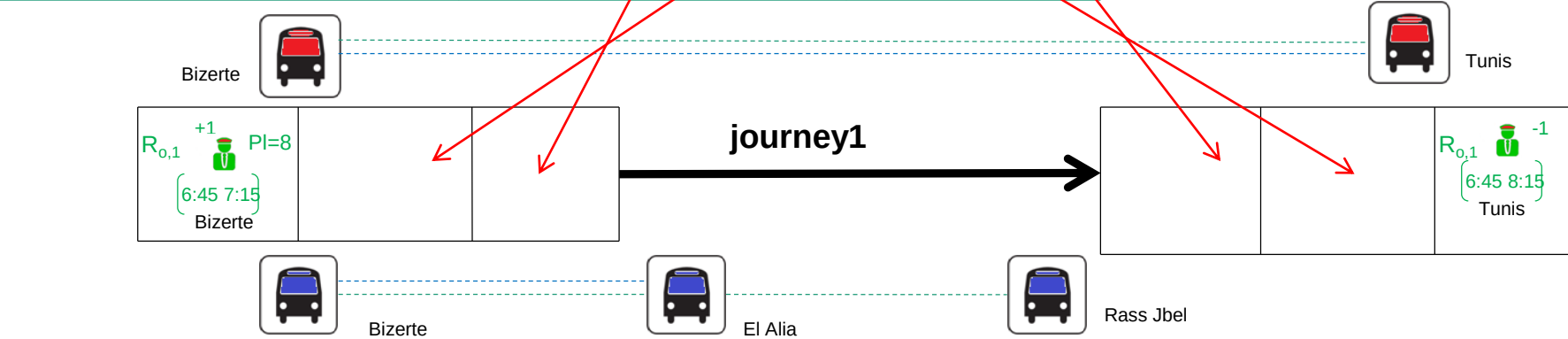
$R_{d,8}$ $\begin{pmatrix} +4 & -4 \\ 6:20 & 7:20 \end{pmatrix}$ Bizerte Tunis	$R_{d,7}$ $\begin{pmatrix} +2 & -2 \\ 6:20 & 7:20 \end{pmatrix}$ Bizerte Tunis	$R_{d,6}$ $\begin{pmatrix} +30 & -30 \\ 6:45 & 7:30 \end{pmatrix}$ Bizerte El Alia	$R_{d,3}$ $\begin{pmatrix} +3 & -3 \\ 6:15 & 7:15 \end{pmatrix}$ Bizerte Tunis	$R_{d,2}$ $\begin{pmatrix} +4 & -4 \\ 6:30 & 7:45 \end{pmatrix}$ Bizerte Tunis
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Request demand



$R_{o,5}$ $\begin{pmatrix} +1 & 0 & -1 \\ 6:30 & 7:30 \end{pmatrix}$ Bizerte El Alia Rass Jbel PI=30	$R_{o,4}$ $\begin{pmatrix} +3 & -2 \\ 6:45 & 7:30 \end{pmatrix}$ Bizerte Rass Jbel PI=4
---	---

Request offer

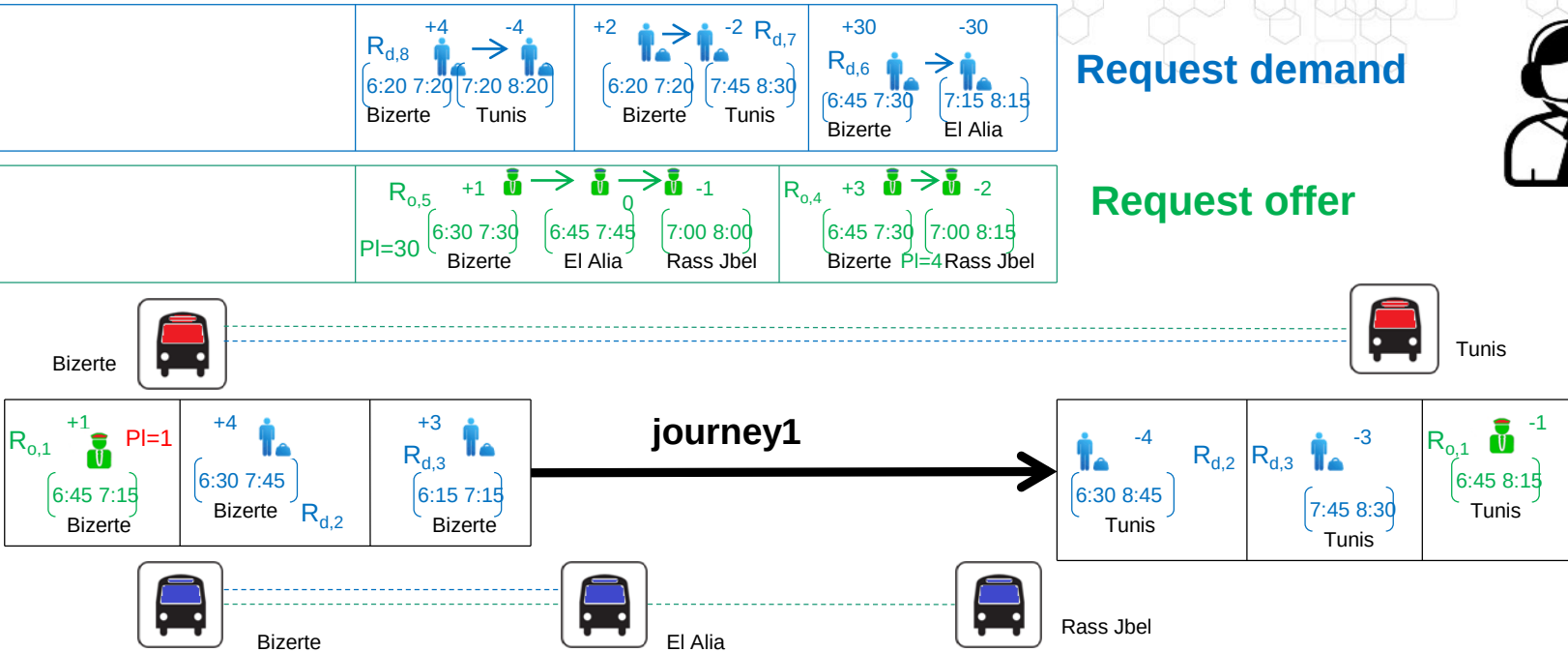


Conclusions and perspectives

Request demand



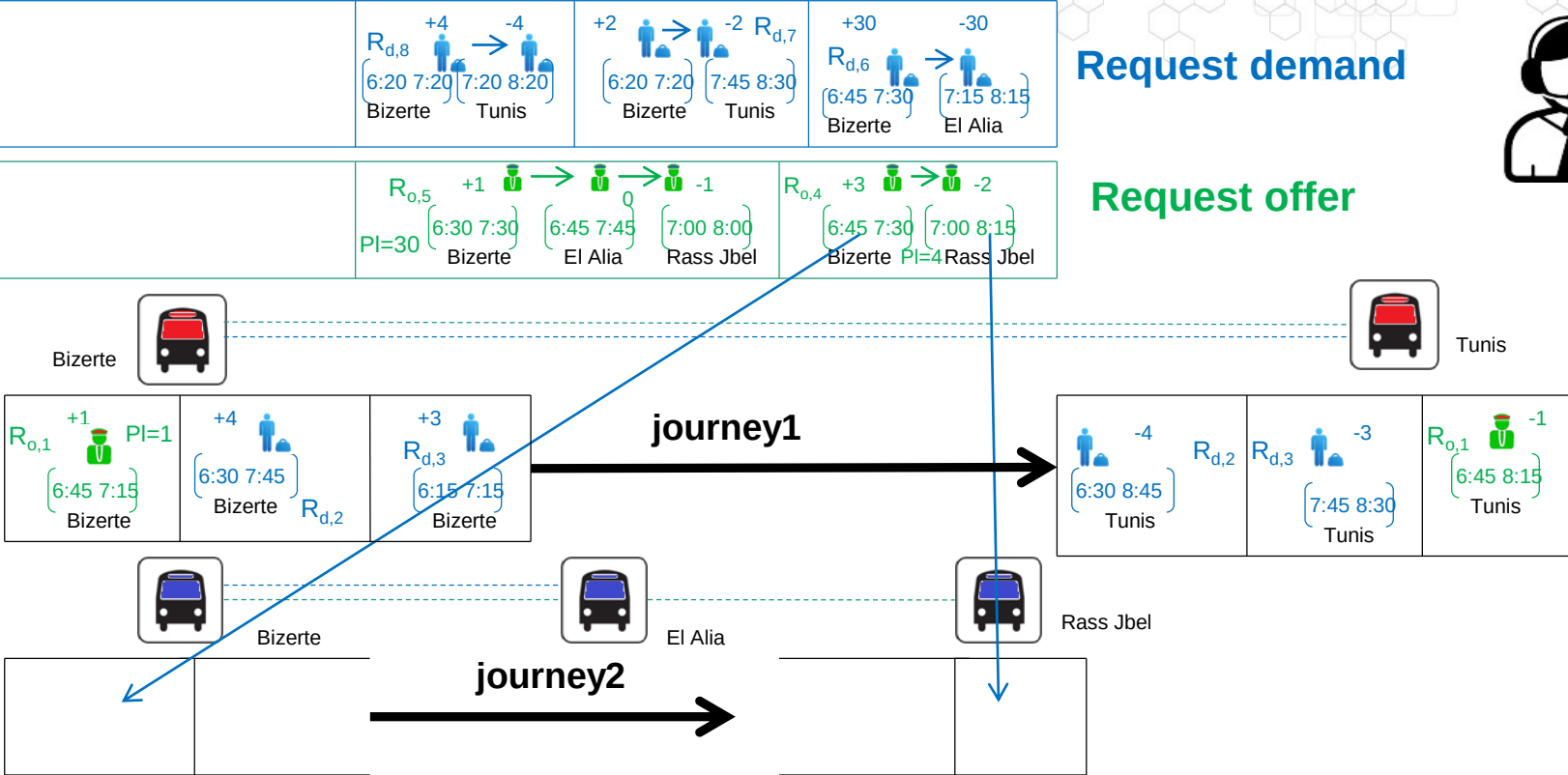
Request offer



Request demand



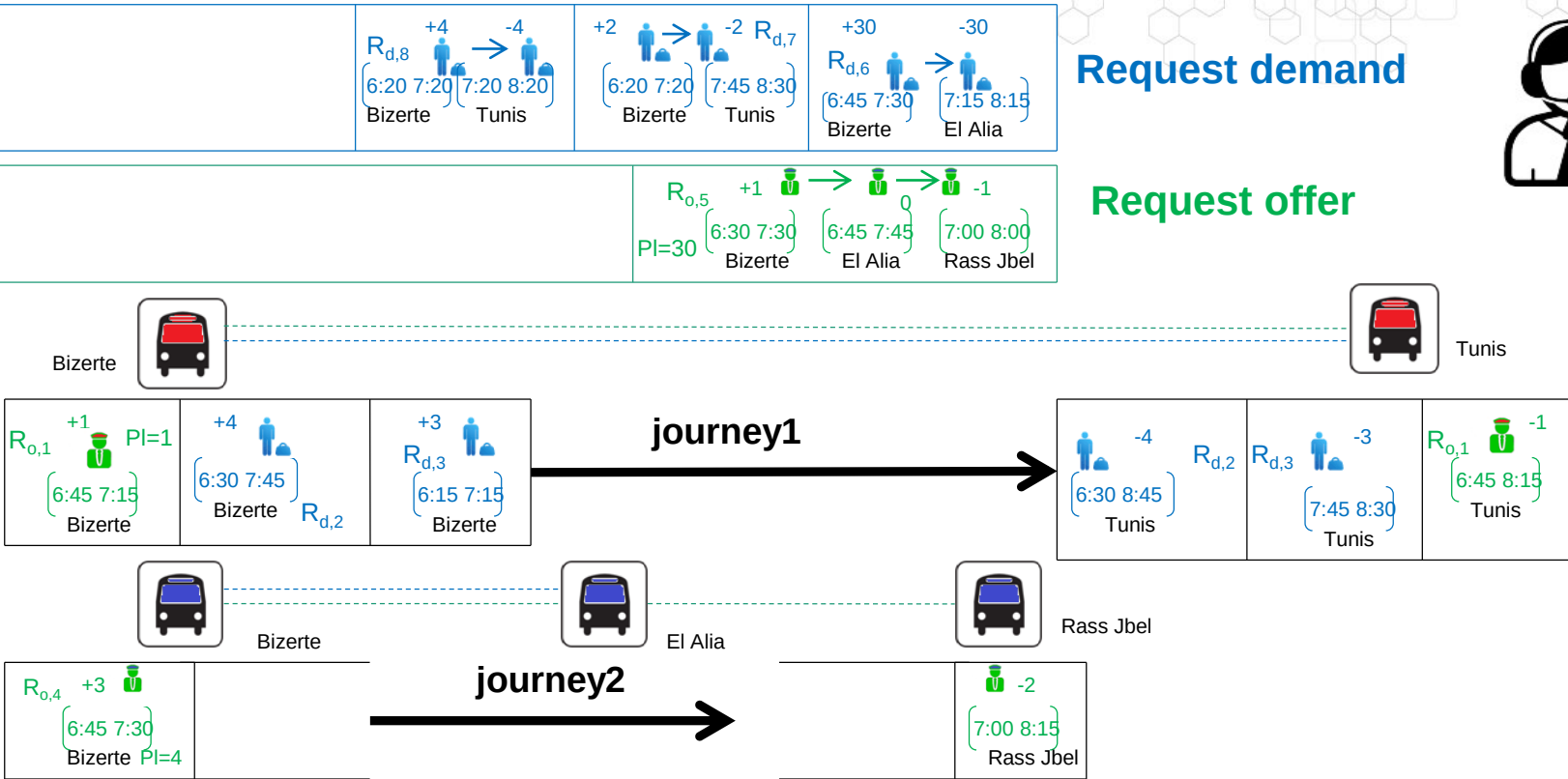
Request offer



Request demand



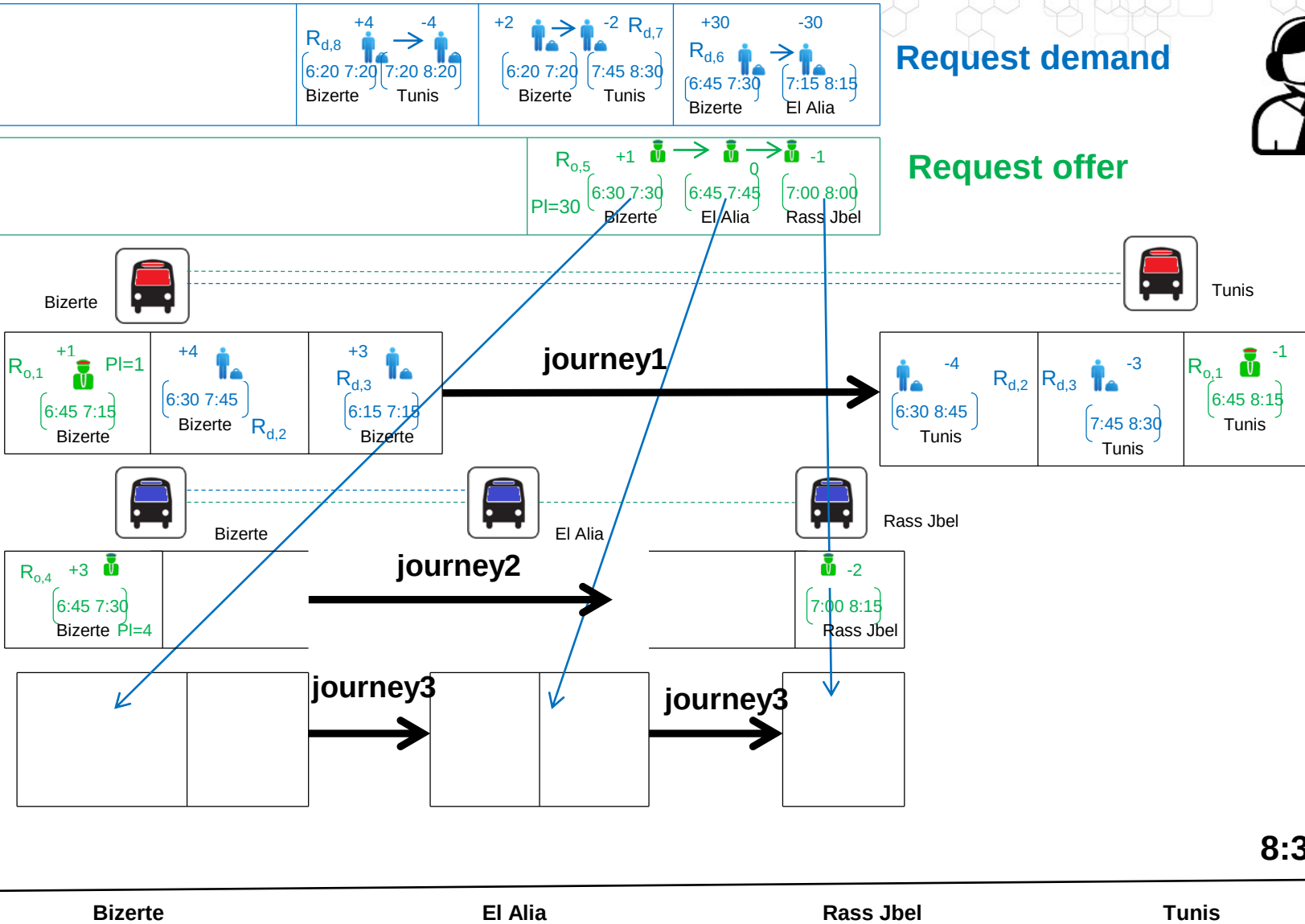
Request offer



Request demand

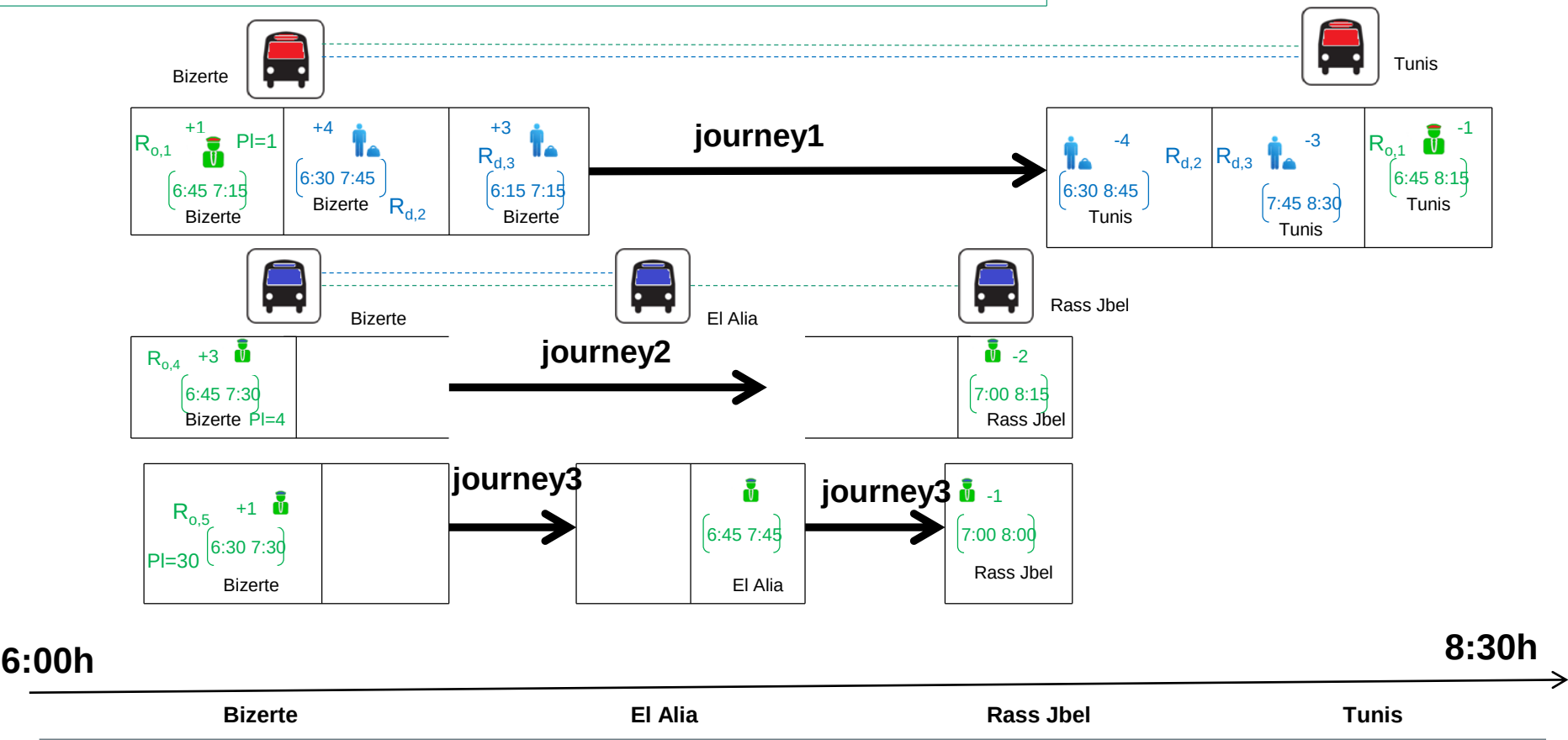


Request offer



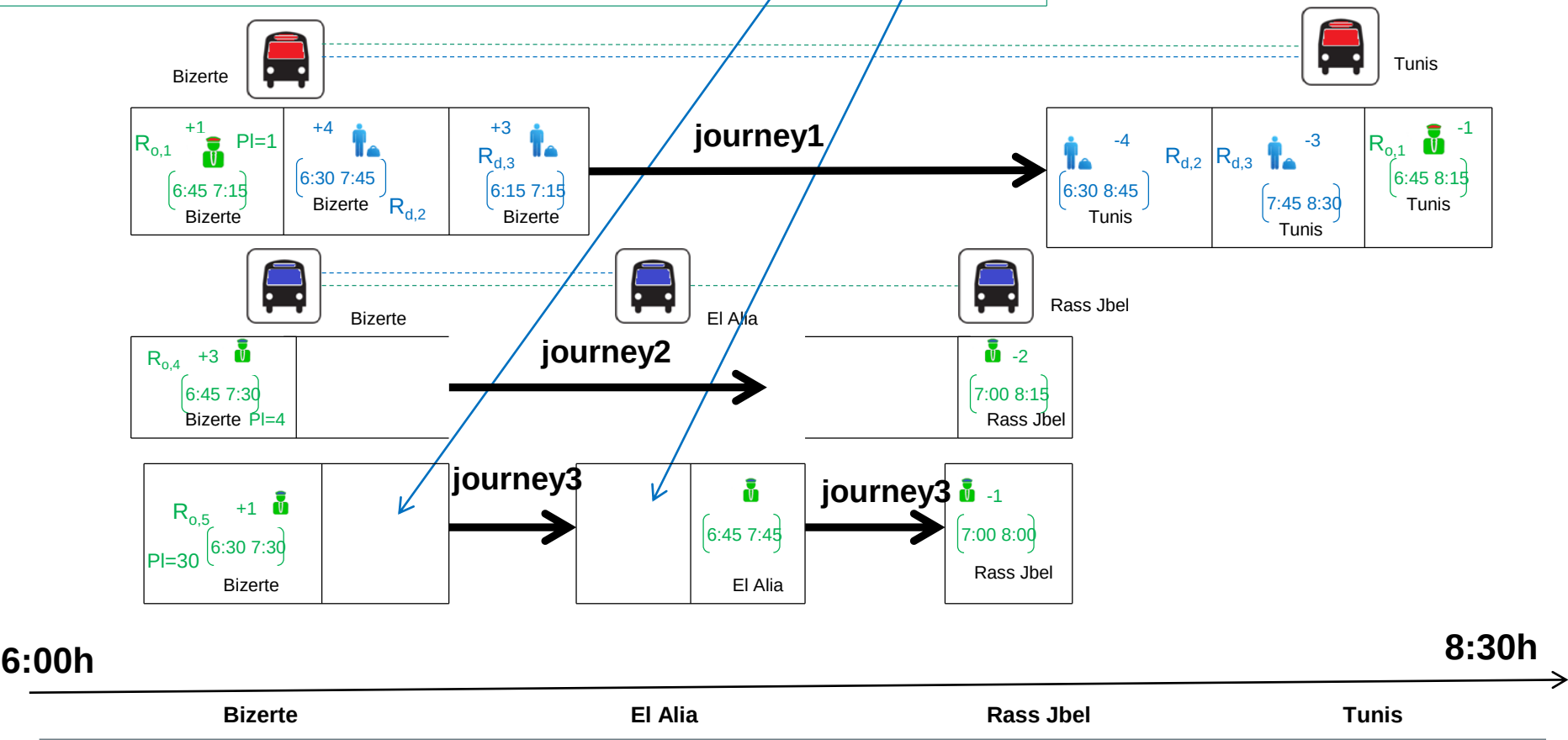
Request demand

Request offer



Request demand

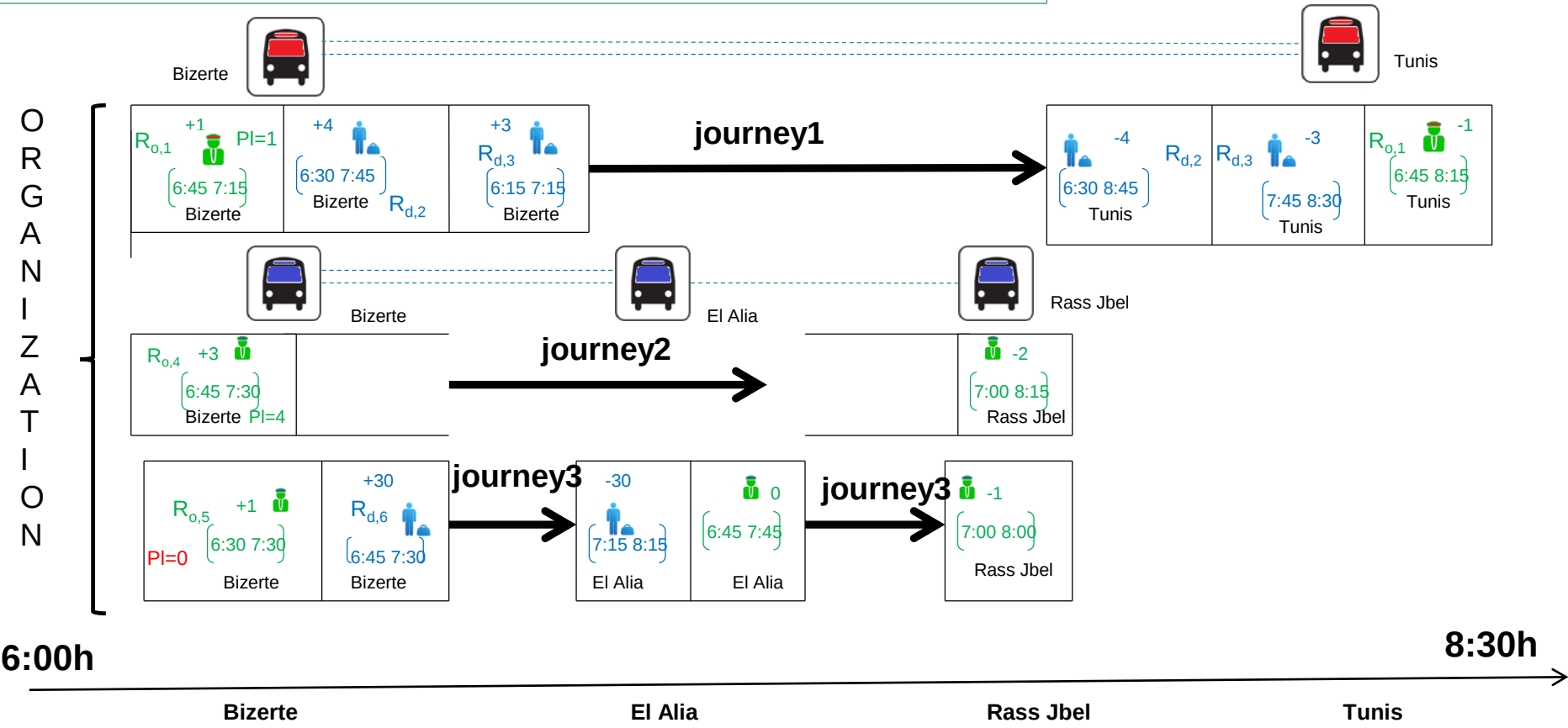
Request offer



Request demand



Request offer



Multi constraints

✚ The Louage's system makes numerous actors to interact : drivers, passengers, customers, operators (central operator vs decentralized operators).

Constraints are classified in several groups :

1. Coherence constraints and normalization of time windows in each request.
 - DTW Normalization into each itinerary of offer request.
 - TW coherence between each position in a request and the corresponding position in of the itinerary computed
2. Louage' s lines constraints : authorized, declared during louage process, etc.
3. Itineraries' constraints
4. Stop position dependencies respected between request and organization(s)
5. Requests orders respected between submission to operators and rank into each organization computed.
6. Constraints on state transitions.
7. Capacity constraints.
8. Price constraints.

Louage lines constraints

✚ The set of **Ligne_c** associated to a driver c.

$$Lignes_c = \{Ligne_{c,l}\} \quad \text{with} \quad Ligne_{c,l} = Pos_{C,c,l,1}[h_1^-, h_1^+], \dots, Pos_{C,c,l,p}[h_p^-, h_p^+], \dots, \\ Pos_{C,c,l,\overline{Ligne_{c,l}}}[h_{\overline{Ligne_{c,l}}}^-, h_{\overline{Ligne_{c,l}}}^+]$$

✚ The set of **Lignes** of overall system.

$$Lignes = \{Ligne_{nl \in [1, \overline{Lignes}]}\} = \bigcup_{c=1}^{\overline{C}} \left(\bigcup_{l=1}^{\overline{Ligne_c}} Ligne_{c,l} \right)$$

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Constraints upon lines of a driver :

✚ not have to exceed **NbMaxLpC = 3**.

$$\forall \quad c \in [1, \overline{C}], \quad NbMaxLpC = 3$$

Constraints linking requests' itineraries with those initialized with lines

For an offer request

$\prod_B(A)$ defines the projection of A upon B. For an initialized request, the projection of an itinerary and its associated positions and TW correspond with positions, and TW, of an existing line in **Lignes**.

$$\forall R_{o,i \in [1, \overline{R_o}]} \text{ et } R_{o,i}.state = "initialisée", \exists nl \in [1, \overline{Ligne}]$$

$$\prod_{pos_{o,p}[h_p^-, h_p^+]} (Itin_{o,i}) = Ligne_{nl}$$

For a demand request

✚ For such request, the itinerary projection and its associated positions and TW correspond but relaxed (greater or equal) to positions and TW of an existing line in **Lignes**.

$$\forall R_{d,j \in [1, \overline{R_d}]} \text{ et } R_{d,j}.state = "initialisée", \exists nl \in [1, \overline{Ligne}]$$

$$\prod_{pos_{d,p}[h_p^-, h_p^+]} (Itin_{d,j}) \subseteq Ligne_{nl}$$

Multi objectives

✚ Louage is intrinsically multi-objectives. As each actor (operator, driver, customer) has its own view, operational objectives may be different from those expressed in academic works :

1. maximization of occupation rate (of vehicles)
2. maximization of contractualized requests
3. respectfull to FIFO list of requests
4. minimization of standart deviation of waiting time
5. minimization of the overall sum of running itineraries duration
6. maximisation of drivers earnings

Maximize the number of requests contractualized

✚ that remains maximizing $\overline{R_{o,con}}$ and $\overline{R_{d,con}}$
and then minimizing the number of requests non contractualized $\overline{R_{o,!con}}$ et $\overline{R_{d,!con}}$

✚ with respect to FIFO list of requests.

Minimize the global duration TT (Total Time) of journeys (of an organization)

$$\forall Org_{Cse \in [1, \overline{Orgs}]} \in Orgs,$$

$$\forall Itin_{Cse, k \in [1, \overline{Org_{Cse}}]} = Pos_{Cse, k, 1}, \dots, Pos_{Cse, k, p}, \dots, Pos_{Cse, k, \overline{Itin_{Cse, k}}}$$

$$TT = \sum_{Cse = Cse_1}^{\overline{Cse_{Org_{Cse}}}} \sum_{p=1}^{\overline{Itin_{Cse, k}}} M_{o-d}(Pos_{Cse, k, p}, Pos_{Cse, k, p+1}).t$$

Associated mono-objective

- ✚ If in the formalization of Louage's problem, we adopt multi-objectives,
- ✚ our proposed algorithm **GILA (Greedy Incremental Louage's Algorithm)** implements a mono objective:
 - ✚ maximizing the number of contractualized requests respectfull to registration order.
- ✚ **GILA** generates then a single organization at each iteration of GILA's call.

Alaorithm description

Algorithm 1 Greedy louage's incremental algorithm($Offers_{init}$ O_{init} , $Demands_{init}$ D_{init} ,
Organization Org)

```

1: //validation des listes des requêtes offre et des requêtes demande initialisées
2:  $O_v \leftarrow ValidateInitializedRequests(O_i)$  //contrainte 3.2
3: //validateInitializedRequests est une fonction qui assure la validation d'une liste de requêtes
  offre ou des requêtes demande
4:  $D_v \leftarrow ValidateInitializedRequests(D_i)$  //contrainte 3.2
5: //initialisation de la liste des organisations
6: if  $Org! = Null$  then
7:    $Orgs = Concat(Orgs, Org)$ 
8: else
9:    $Orgs = new Organization()$ 
10: end if
11: //ajout des offres validées dans la premiere organisation à la suite des courses de l'organisa-
  tion existante
12:  $Orgs.Org_1 \leftarrow Concat(Org, O_v)$ 
13:  $PcourseOrgs \leftarrow Orgs$ 
14:  $PitinOrgs \leftarrow PcourseOrgs$ 
15:  $PpositinOrgs \leftarrow PitinOrgs$ 
16:  $PD_v \leftarrow D_v$ 
17: while  $PcourseOrgs! = Null$  &&  $!(PcourseOrgs.état = "contractualisée")$  do
18:   while  $PD_v! = Null$  &&  $!(PD_v.état.equals("contractualisée"))$  do
19:      $PPosD_v \leftarrow D_v$ 
20:     while  $PPosD_v! = Null$  &&  $PpositinOrgs! = Null$  do
21:       if  $IsInserted(PPosD_v, PpositinOrgs, PpositinOrgs.next())$  then
22:         //fonction qui vérifie l'insertion avec respect des contrainte 3.12, 3.13, 3.21, 3.22//
23:          $ValidateLocalTimeWindow(PPosD_v, PpositinOrgs.next())$ 
24:          $PropagationOfTimeWindow(PPosD_v, PpositinOrgs, PpositinOrgs.next())$ 
25:          $Update(PPosD_v, PpositinOrgs, PpositinOrgs.next())$ 
26:         //fonction qui vérifie met à jour l'itinéraire avec respect des contraintes 3.12 et
          3.13//
27:          $PpositinOrgs \leftarrow PPosD_v$ 
28:          $PPosD_v \leftarrow PPosD_v.next()$ 
29:       else if  $PPosD_v = Null$  then
30:         //une demande a été injecté

```

```

31:    $D_{con} \leftarrow D_{con} + PD_v$ 
32:    $D_v \leftarrow D_v - PD_v$ 
33:    $PpositinOrgs.PlaceO = PpositinOrgs.PlaceO + PD_v.NP$ 
34:    $PpositinOrgs.PlaceL = PpositinOrgs.PlaceL - PD_v.NP$ 
35:    $PD_v \leftarrow PD_v.next()$ 
36:    $PPosD_v \leftarrow PD_v$ 
37: else
38:   //insertion impossible et avancement du pointeur sur la prochaine requête demande
39:    $PD_v \leftarrow PD_v.next()$ 
40:    $PPosD_v \leftarrow PD_v$ 
41: end if
42: end while
43:    $PD_v \leftarrow PD_v.next()$ 
44: end while
45:  $PCourseOrgs \leftarrow PCourseOrgs.next()$ 
46:  $PitinOrgs \leftarrow PCourseOrgs$ 
47:  $PpositinOrgs \leftarrow PitinOrgs$ 
48: end while
49: return  $Orgs$ 

```

Algorithm 2 ValidateLocalTimeWindow(POS $PPosD_v, POS$ $PpositinOrgs.next()$)

```

1: if  $PPosD_v.h^+ + M_{o-d}(PPosD_v, PpositinOrgs.next()).t \geq PpositinOrgs.next().h^+$  then
2:    $PPosD_v.h^+ \leftarrow PpositinOrgs.next().h^+ - M_{o-d}(PPosD_v, PpositinOrgs.next()).t$ 
3: end if
4: if  $PPosD_v.h^- + M_{o-d}(PPosD_v, PpositinOrgs.next()).t \geq PpositinOrgs.next().h^-$  then
5:    $PpositinOrgs.next().h^- \leftarrow PPosD_v.h^- + M_{o-d}(PPosD_v, PpositinOrgs.next()).t$ 
6: end if

```

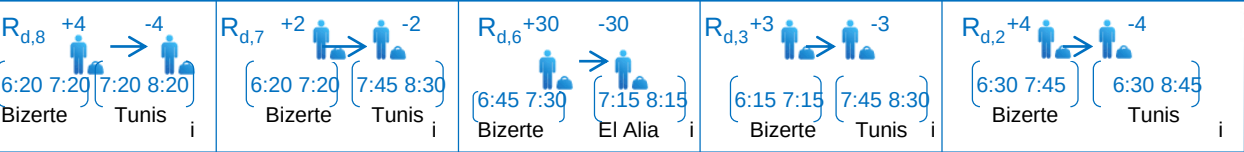
Algorithm 3 PropagationOfTimeWindow(POS $PPosD_v, POS$ $PpositinOrgs, POS$ $PpositinOrgs.next()$)

```

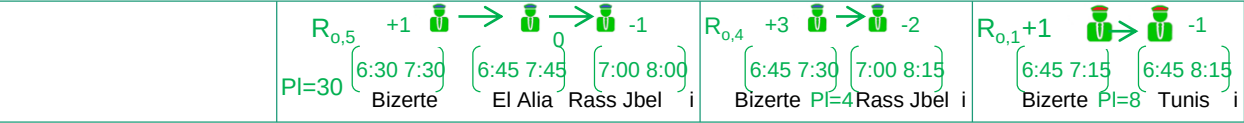
1: if  $PPosD_v.h^- + M_{o-d}(PPosD_v, PpositinOrgs.next()).t \geq PpositinOrgs.next().h^-$  then
2:   /*Propagation à droite*/
    $PpositinOrgs.next().h^- \leftarrow PPosD_v.h^- +$ 
      $M_{o-d}(PPosD_v, PpositinOrgs.next()).t$ 
    $PPosD_v.h^+ \leftarrow PpositinOrgs.next().h^+ -$ 
      $M_{o-d}(PPosD_v, PpositinOrgs.next()).t$ 
3: else if  $PPosD_v.h^+ - M_{o-d}(PpositinOrgs, PPosD_v).t \leq PpositinOrgs.h^+$  then
4:   /*Propagation à gauche*/
    $PpositinOrgs.h^+ \leftarrow PPosD_v.h^+ -$ 
      $M_{o-d}(PpositinOrgs, PPosD_v).t$ 
    $PPosD_v.h^- \leftarrow PpositinOrgs.h^- +$ 
      $M_{o-d}(PpositinOrgs, PPosD_v).t$ 
5: end if

```

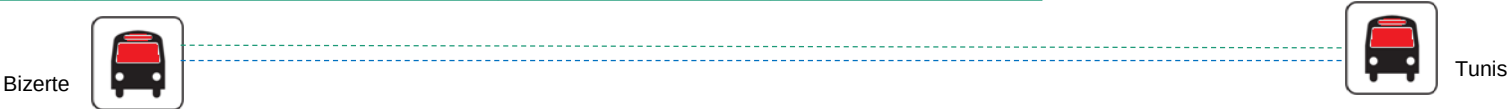
New pairing example when applying GILA



Request demand



Request offer



Legend

i initialized

v validated

c contractualized

pc partially contractualized

Trajet de passager

Trajet de chauffeur

Passager

Chauffeur d'un louage bleu

L'opérateur

Chauffeur d'un louage rouge

Louage à bande bleu

Louage à bande rouge

R_d

R_o

PI

+1

-1

Requête demande

Requête offre

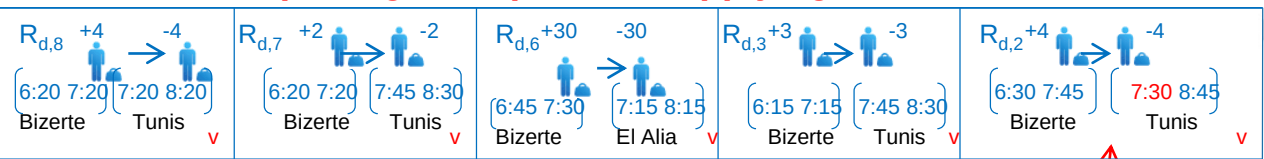
Places libres

Un chauffeur ou un passager montant (PU)

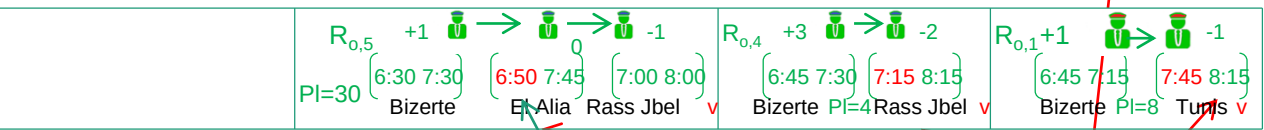
Un chauffeur ou un passager descendant (D)



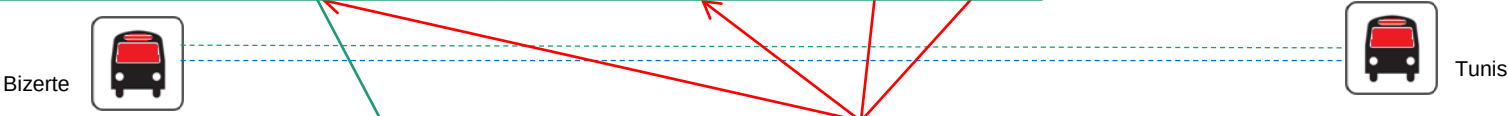
New pairing example when applying GILA



Request demand



Request offer



Normalization of Time Windows (validation of R)
ValidateInitializedRequests()

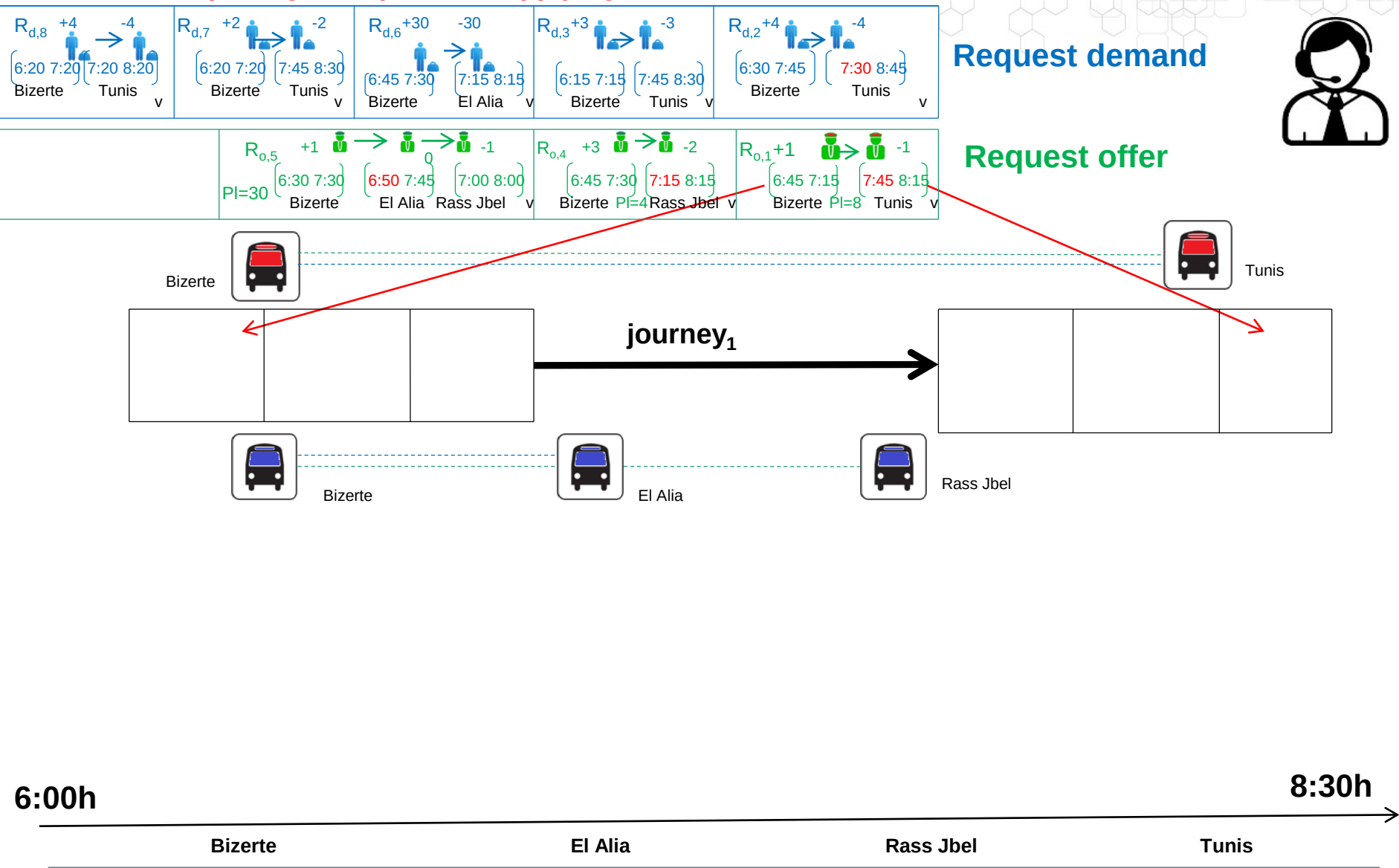


$$Pos_{R_{o,5,1}}.h_v^- = \max(Pos_{R_{o,5,1}}.h_i^-, Pos_{R_{o,5,1}}.h_i^- + d(Pos_{R_{o,5,0}}, Pos_{R_{o,5,1}}), Pos_{R_{o,5,2}}.h_i^- - d(Pos_{R_{o,5,1}}, Pos_{R_{o,5,2}}))$$

Normalization of TW in case of itineraries with vias



New pairing example when applying GILA



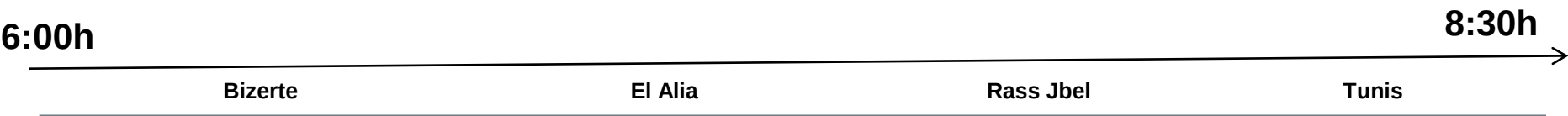
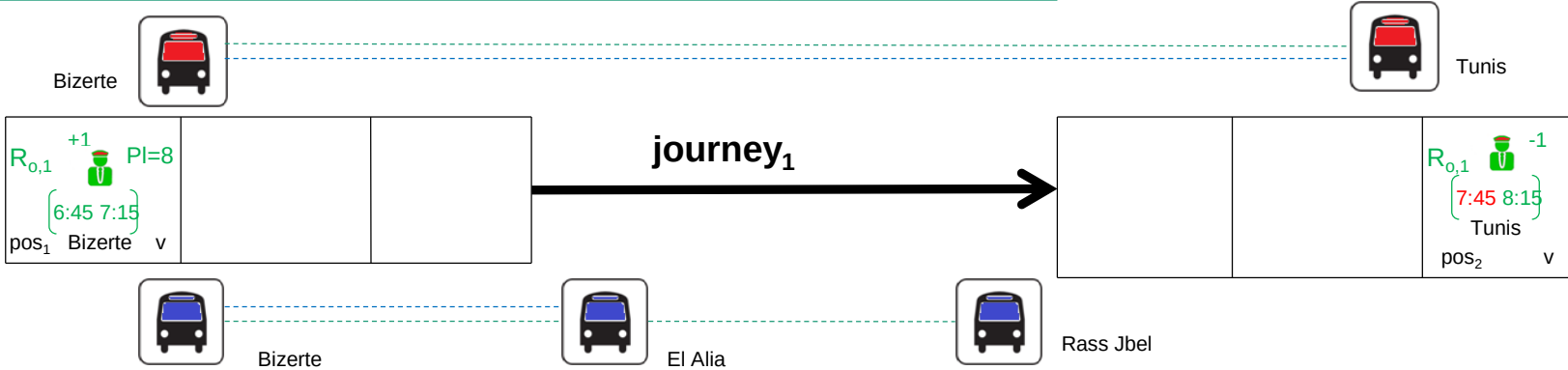
New pairing example when applying GILA

$R_{d,8} +4 \rightarrow -4$ $\left(\begin{smallmatrix} 6:20 & 7:20 \\ \text{Bizerte} & \text{Tunis} \end{smallmatrix} \right)_v$	$R_{d,7} +2 \rightarrow -2$ $\left(\begin{smallmatrix} 6:20 & 7:20 \\ \text{Bizerte} & \text{Tunis} \end{smallmatrix} \right)_v$	$R_{d,6} +30 \rightarrow -30$ $\left(\begin{smallmatrix} 6:45 & 7:30 \\ \text{Bizerte} & \text{El Alia} \end{smallmatrix} \right)_v$	$R_{d,3} +3 \rightarrow -3$ $\left(\begin{smallmatrix} 6:15 & 7:15 \\ \text{Bizerte} & \text{Tunis} \end{smallmatrix} \right)_v$	$R_{d,2} +4 \rightarrow -4$ $\left(\begin{smallmatrix} 6:30 & 7:45 \\ \text{Bizerte} & \text{Tunis} \end{smallmatrix} \right)_v$
$R_{o,5} +1 \rightarrow 0 \rightarrow -1$ $PI=30 \left(\begin{smallmatrix} 6:30 & 7:30 \\ \text{Bizerte} & \text{El Alia} \end{smallmatrix} \right)_v \rightarrow \left(\begin{smallmatrix} 7:50 & 7:45 \\ \text{El Alia} & \text{Rass Jbel} \end{smallmatrix} \right)_v$				
$R_{o,4} +3 \rightarrow -2$ $PI=4 \left(\begin{smallmatrix} 6:45 & 7:30 \\ \text{Bizerte} & \text{Rass Jbel} \end{smallmatrix} \right)_v \rightarrow \left(\begin{smallmatrix} 7:15 & 8:15 \\ \text{Rass Jbel} & \text{Tunis} \end{smallmatrix} \right)_v$				

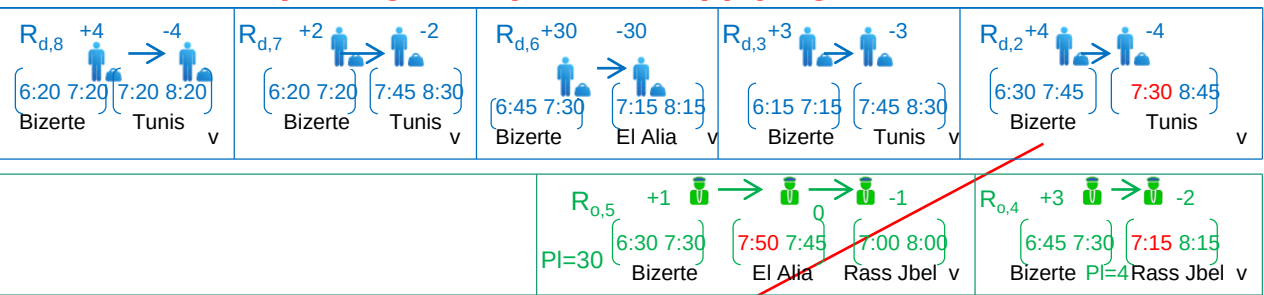
Request demand



Request offer

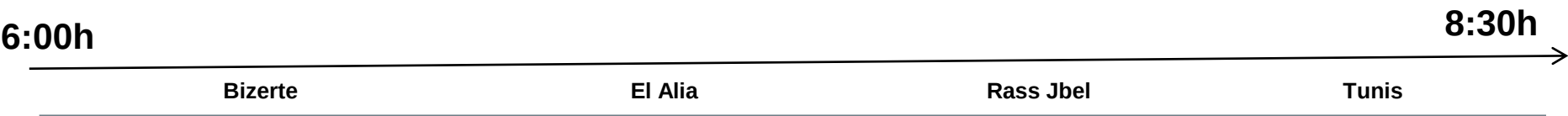
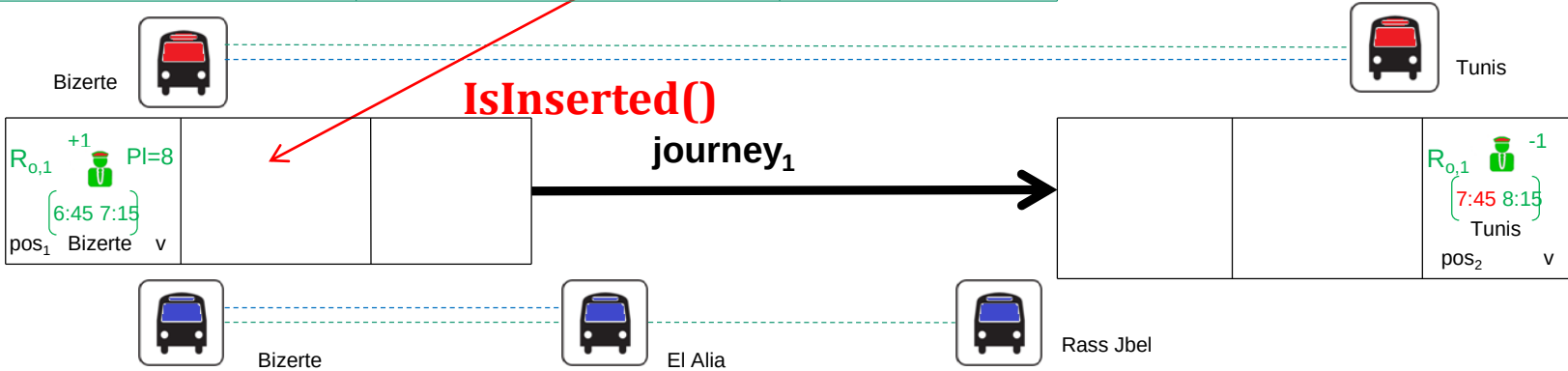


New pairing example when applying GILA

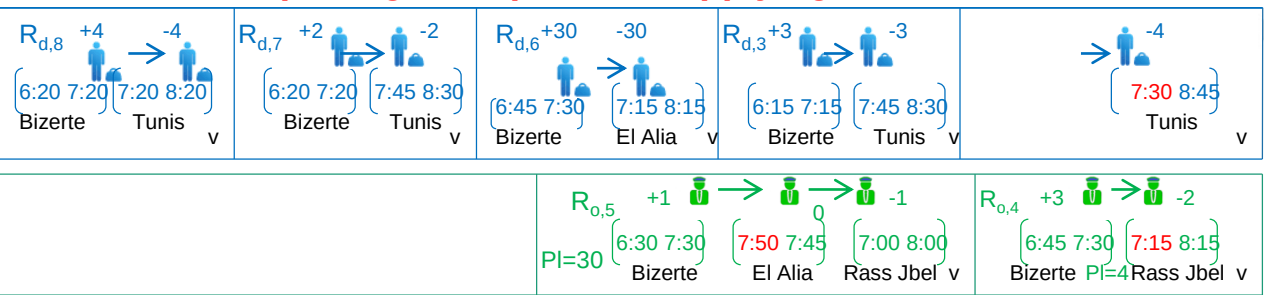


Request demand

Request offer

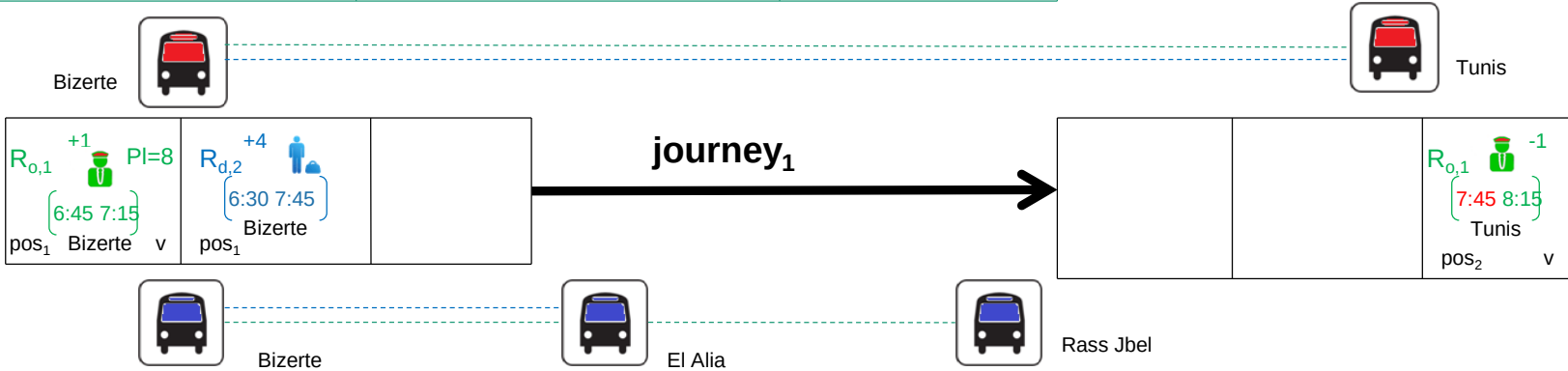


New pairing example when applying GILA



Request demand

Request offer



6:00h

8:30h

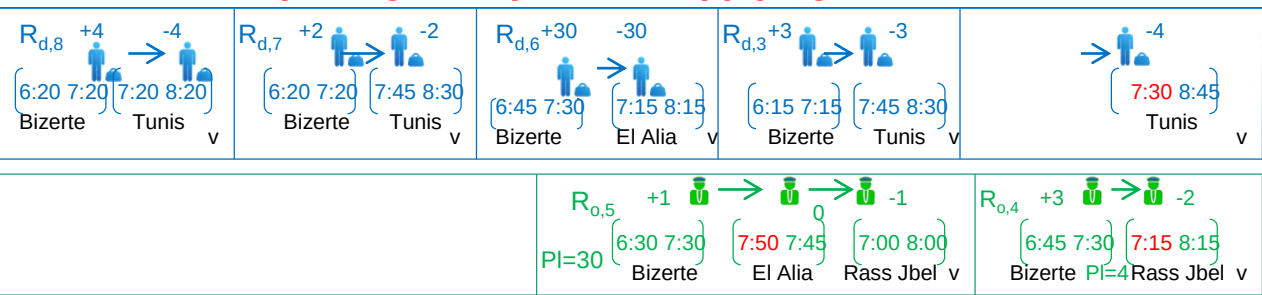
Bizerte

El Alia

Rass Jbel

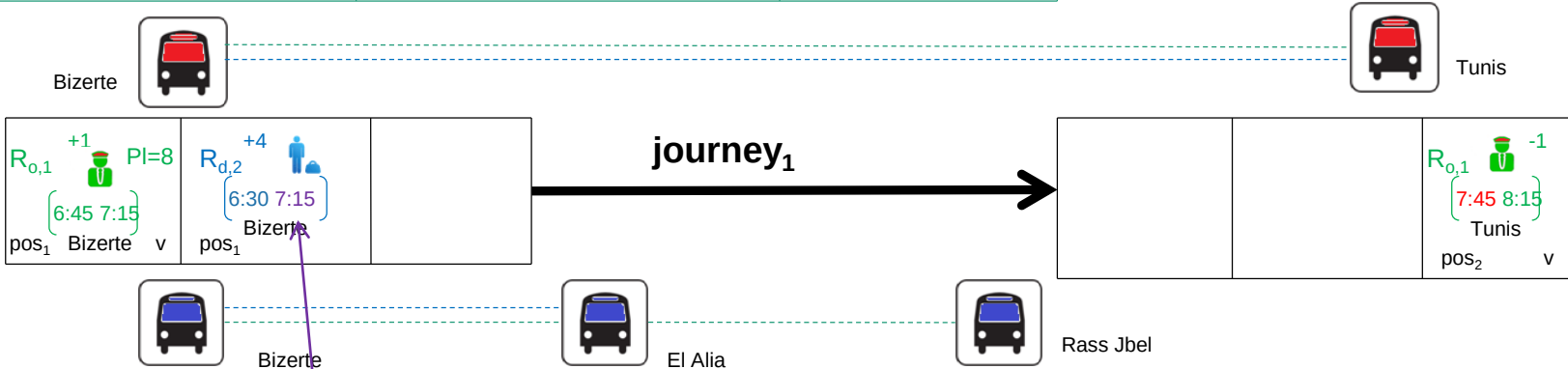
Tunis

New pairing example when applying GILA



Request demand

Request offer



ValidateLocalTimeWindow()

6:00h

8:30h

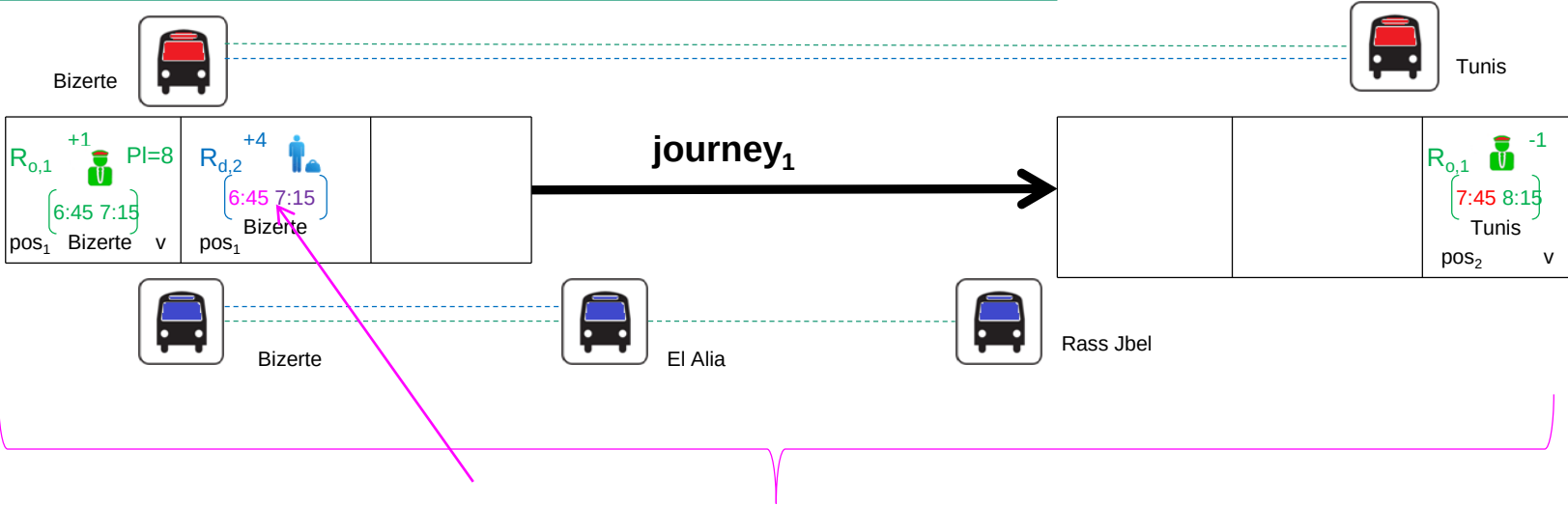
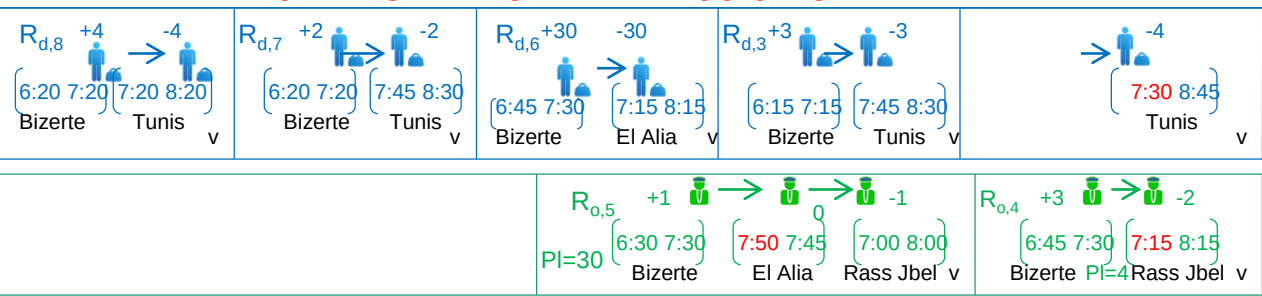
Bizerte

El Alia

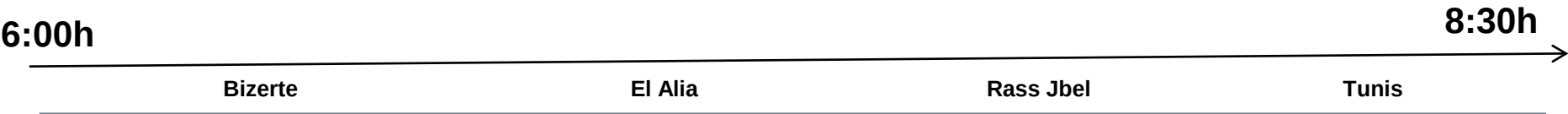
Rass Jbel

Tunis

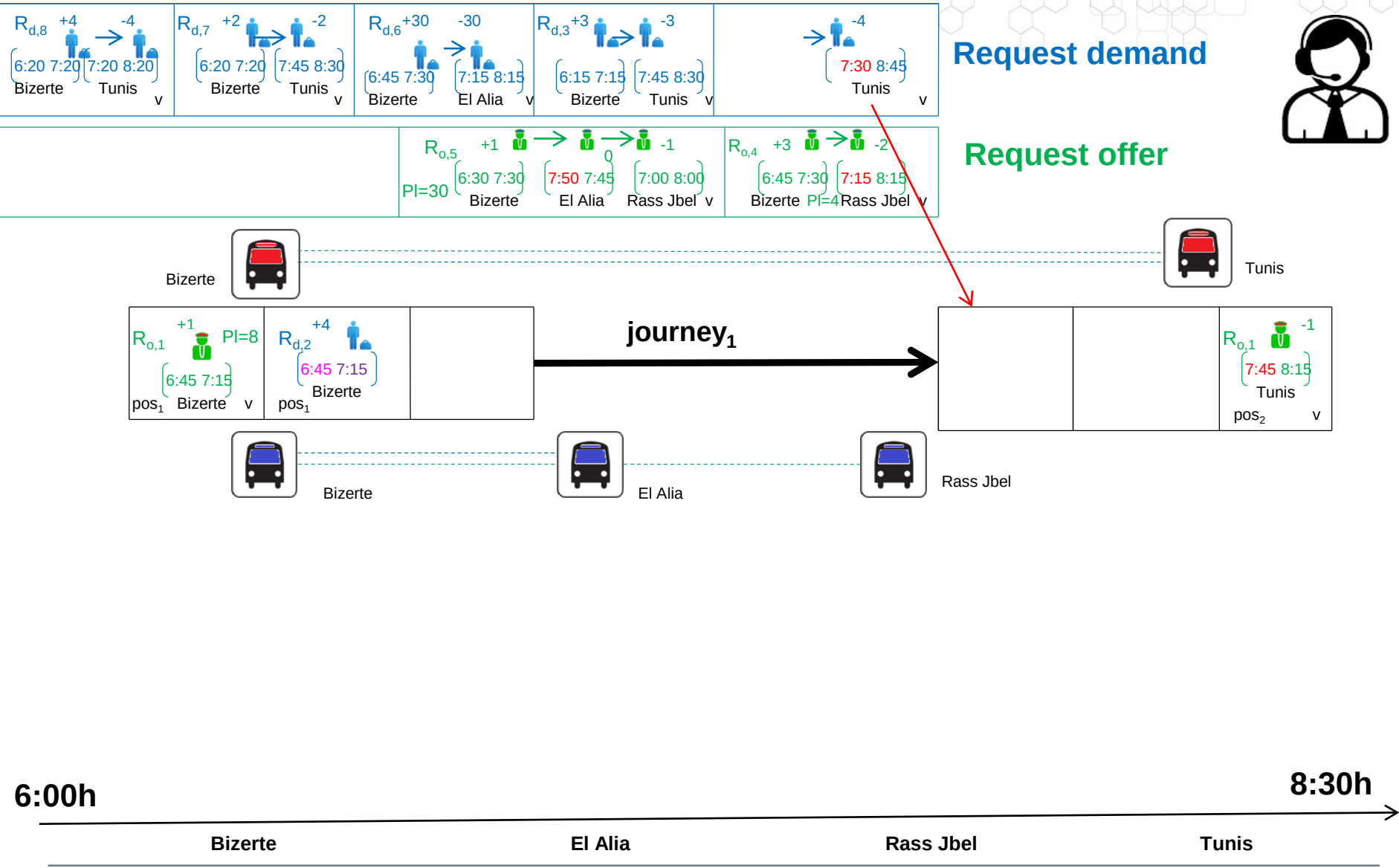
New pairing example when applying GILA



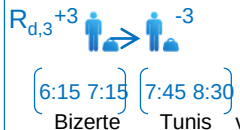
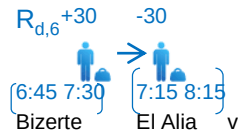
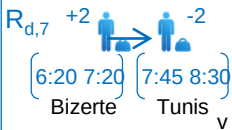
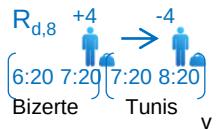
PropagationOfTimeWindow()



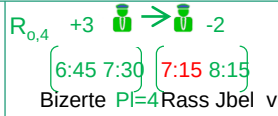
New pairing example when applying GILA



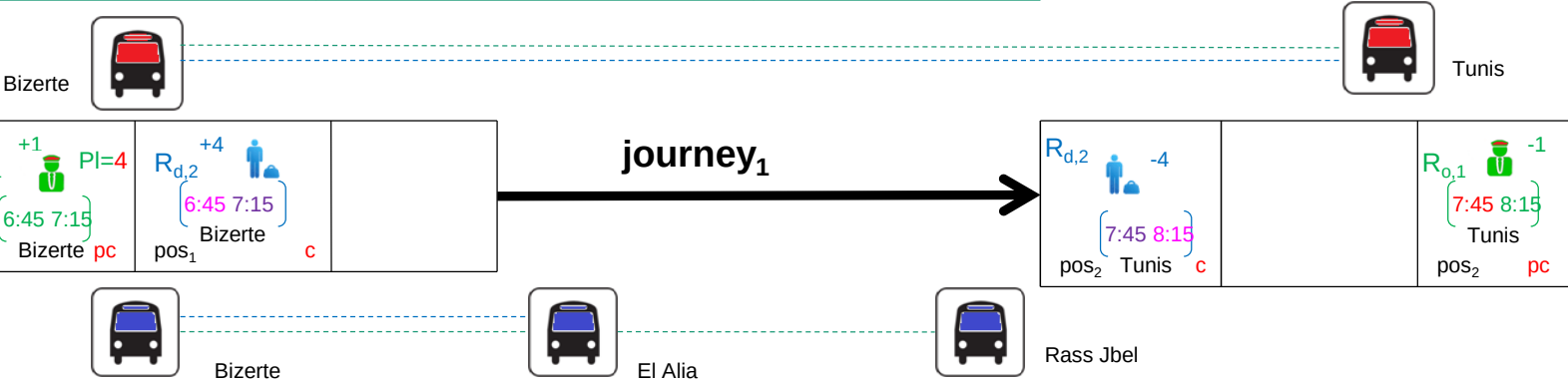
New pairing example when applying GILA



Request demand

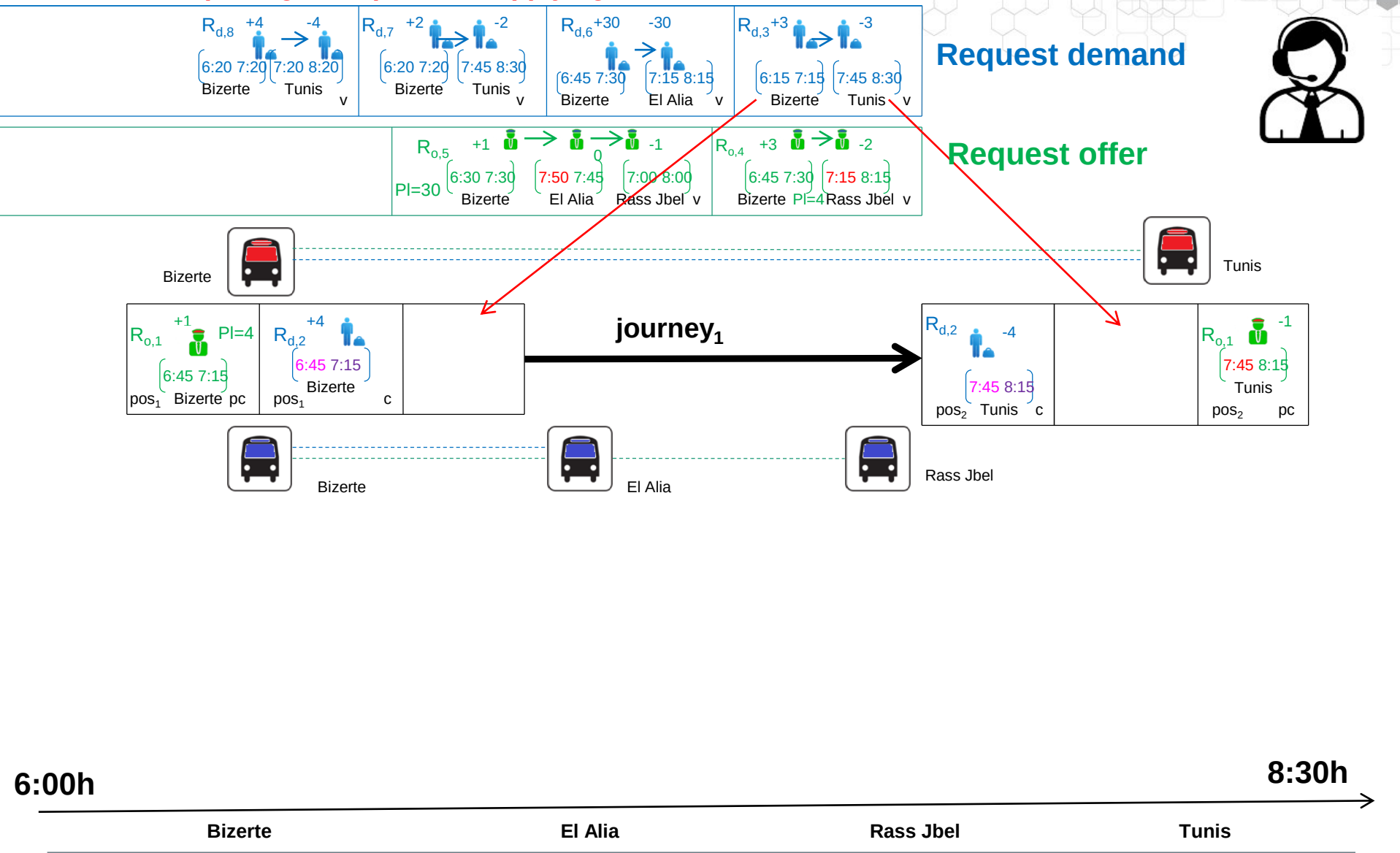


Request offer



Conclusions and perspectives

New pairing example when applying GILA

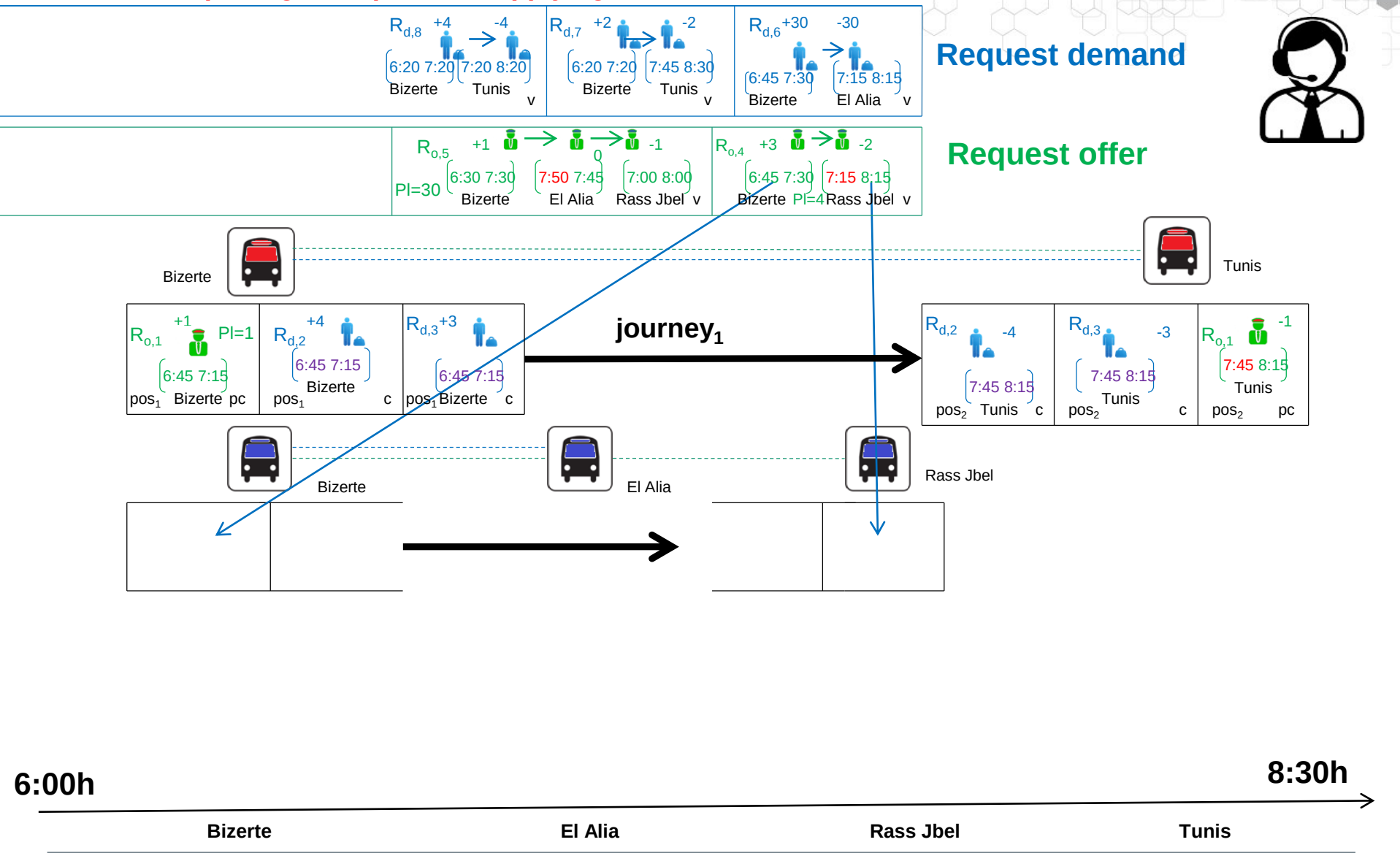




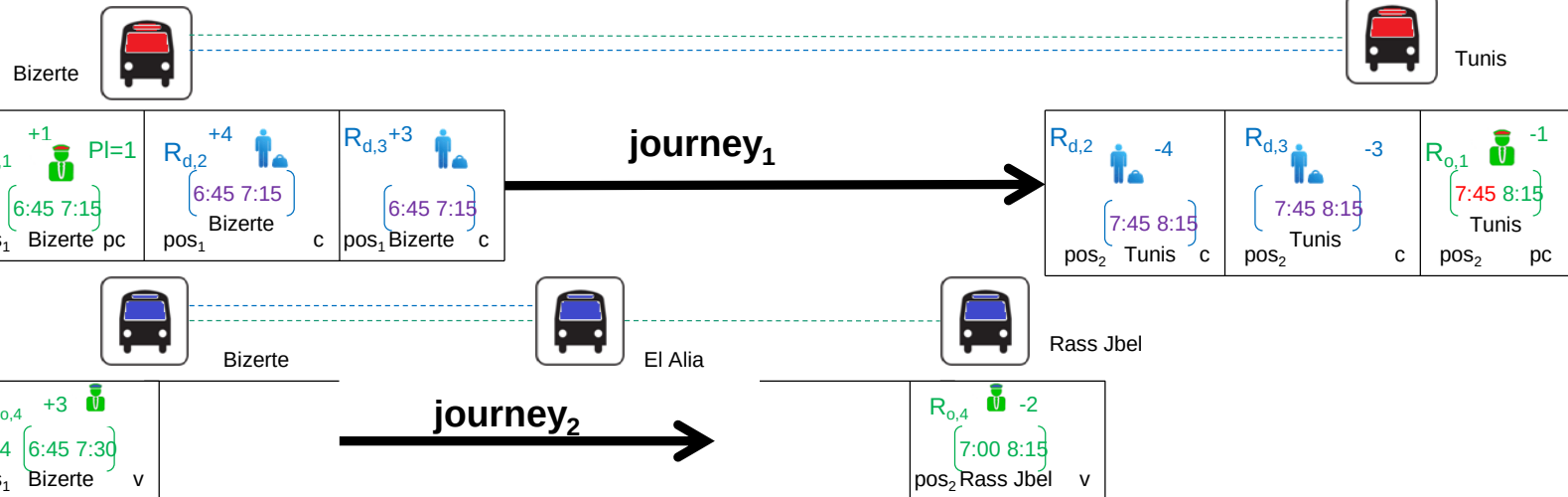
8:30h

Tunis

New pairing example when applying GILA



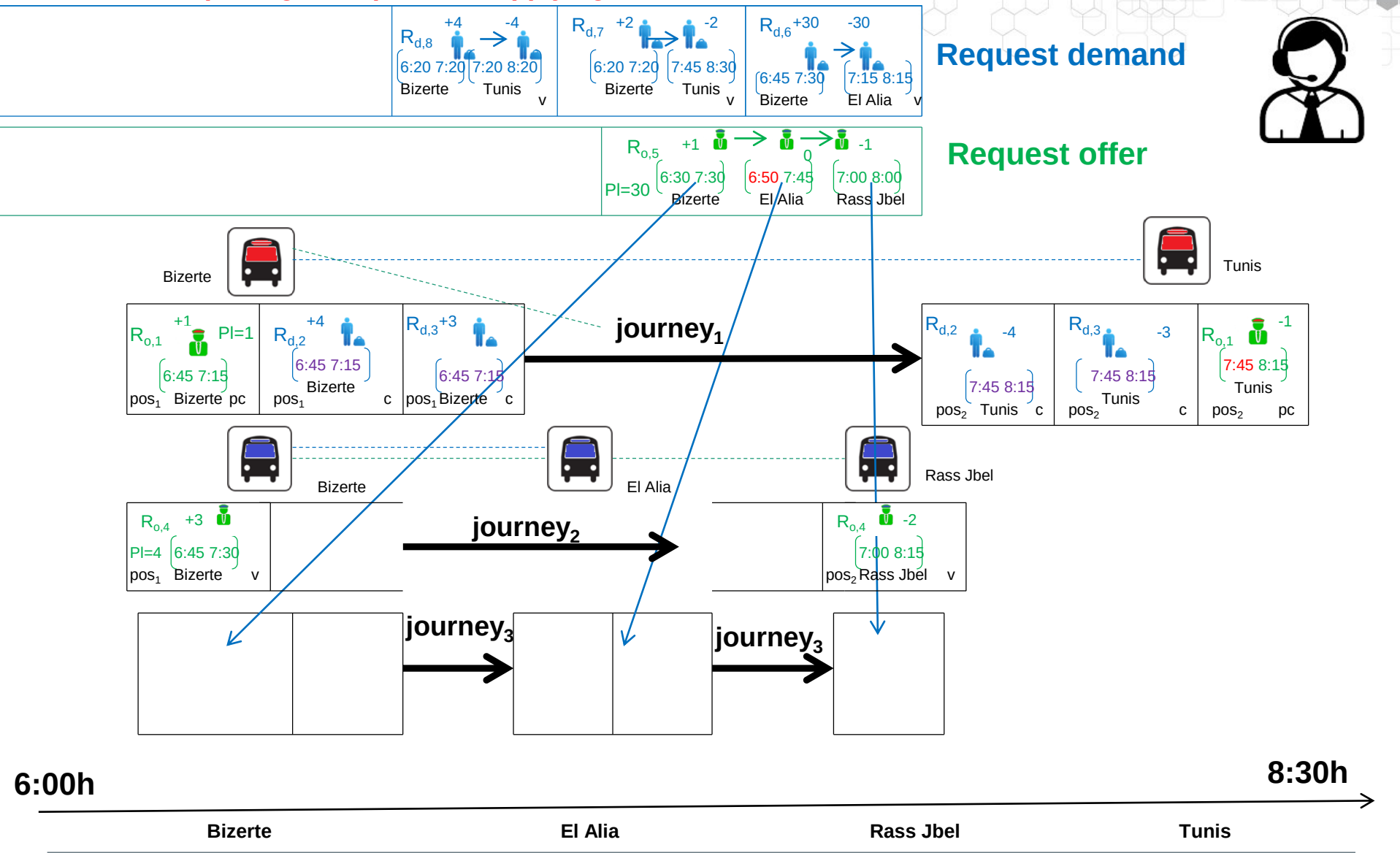
Request demand



8:30h

Tunis

New pairing example when applying GILA

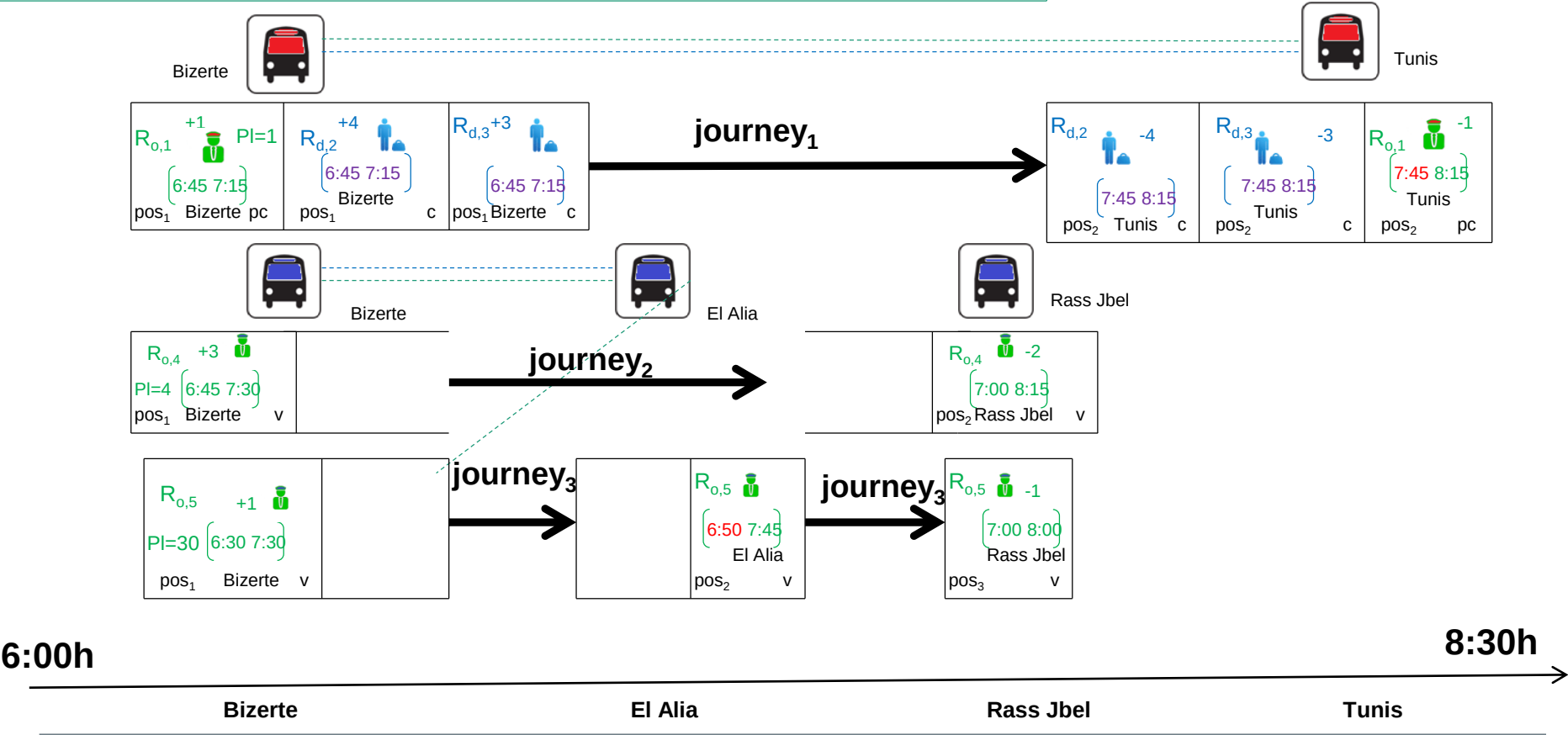


New pairing example when applying GILA

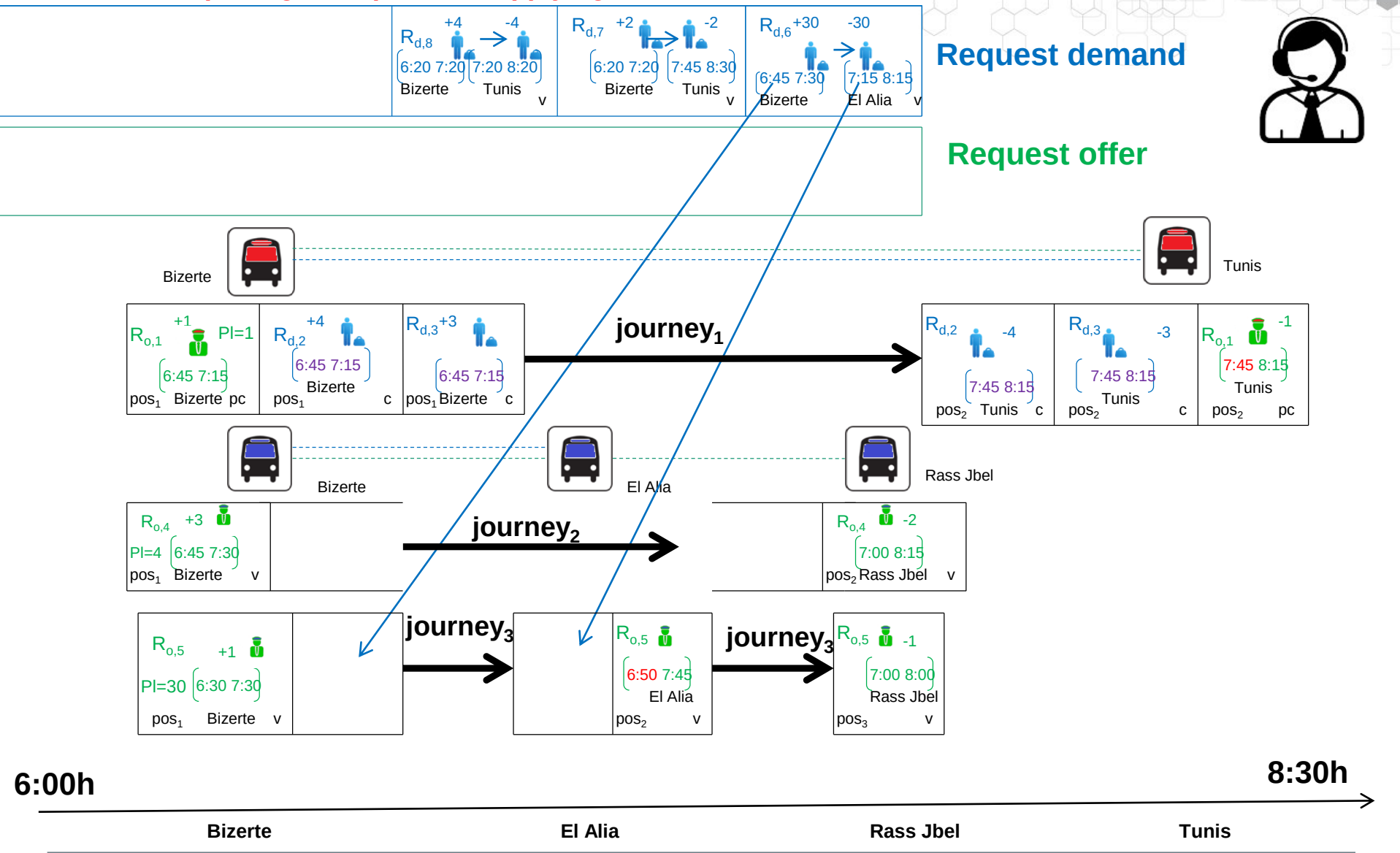
$R_{d,8}$ $\begin{pmatrix} 6:20 & 7:20 \\ \text{Bizerte} & \text{Tunis} \end{pmatrix}$ +4 -4 v	$R_{d,7}$ $\begin{pmatrix} 6:20 & 7:20 \\ \text{Bizerte} & \text{Tunis} \end{pmatrix}$ +2 -2 v	$R_{d,6}$ $\begin{pmatrix} 6:45 & 7:30 \\ \text{Bizerte} & \text{El Alia} \end{pmatrix}$ +30 -30 v
--	--	--

Request demand

Request offer



New pairing example when applying GILA



New pairing example when applying GILA

Request demand



Request offer

ORGANIZATION

Itération 1

6:00h

8:30h

Bizerte

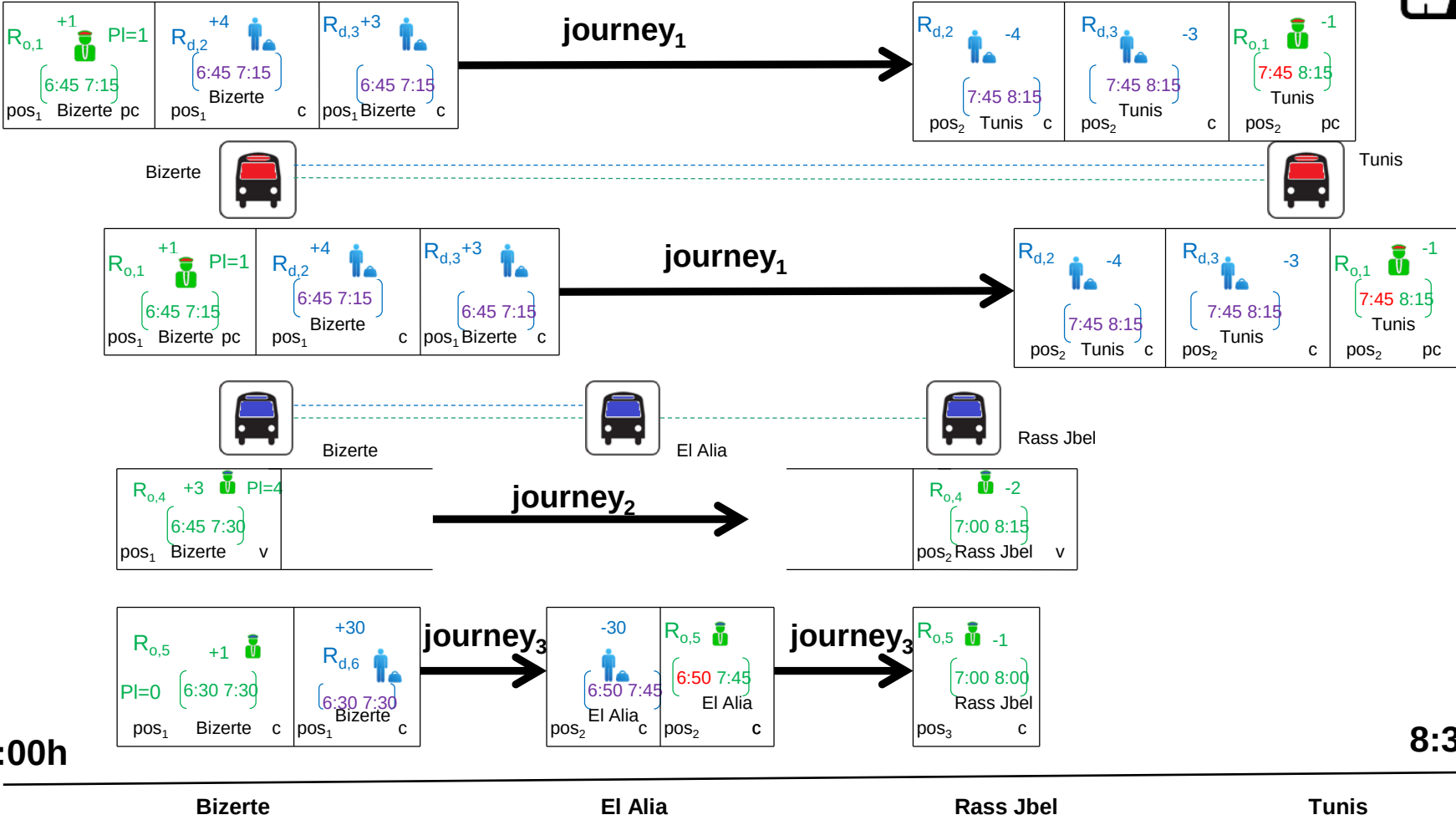
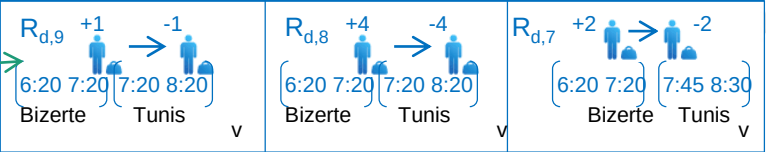
El Alia

Rass Jbel

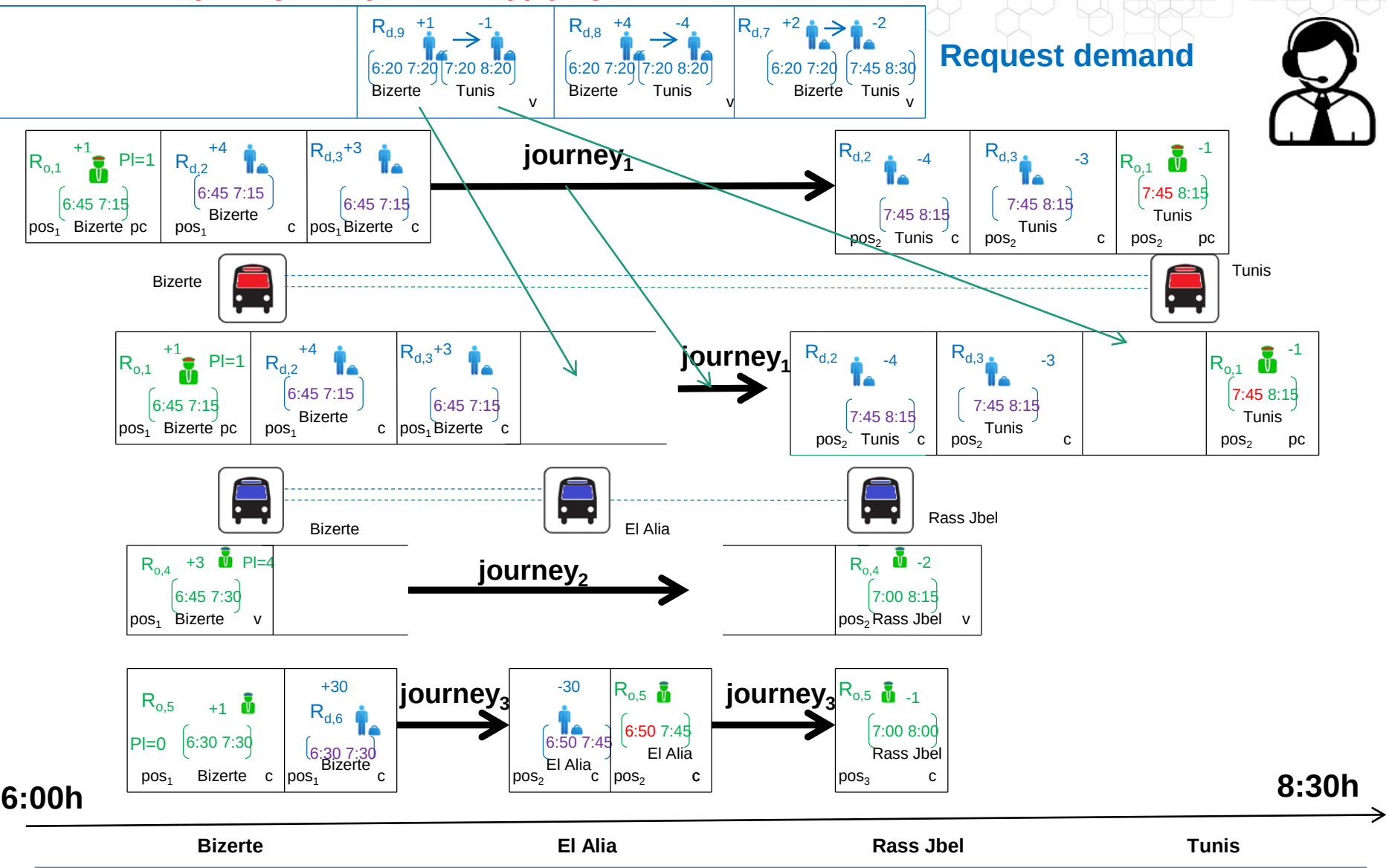
Tunis

New pairing example when applying GILA

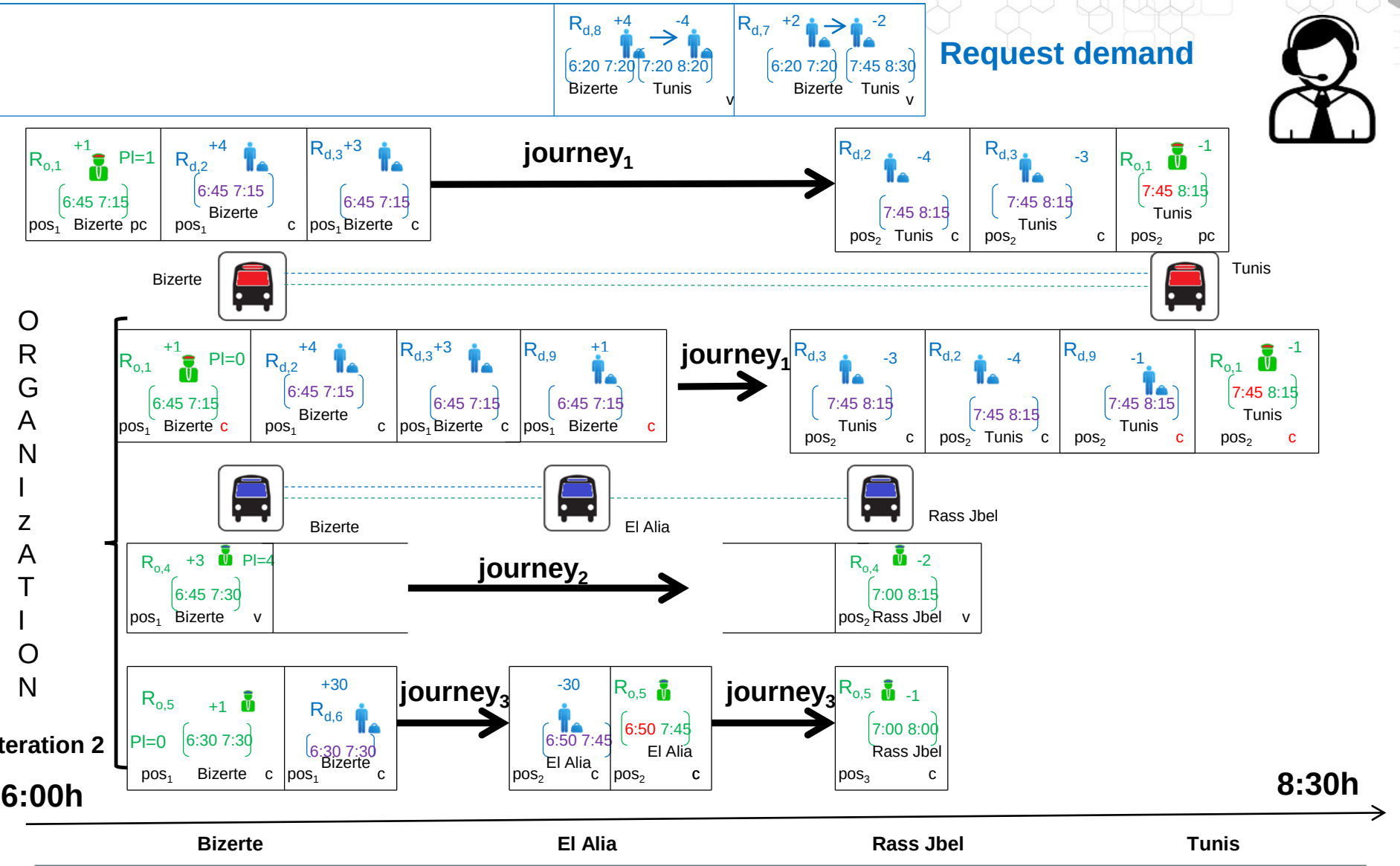
A new request Demand
is being receipted
Iteration 2



New pairing example when applying GILA



New pairing example when applying GILA



Tests

✚ Tests are realized from two files of request offer and request demand. From a 2nd iteration or more, tests involve another file of organization selected.

✚ Structure of a line in a request demand file :
**idrequestdemand, itinerary{posi[earlierdate/latestdate/pickup/delivery]},
nombreofplaces, price, status**

```
requête_demandes - Bloc-notes
Fichier Edition Format Affichage ?

idrequetedemande,itineraire{posi[dateauplustot/dateauplustard/pickup/delivery]},nombredeplaces,montant,etat
4,bizerte/6/30/7/45/+1/0_tunis/6/30/8/45/0/-1,1,4.75,initialise
5,bizerte/6/30/7/45/+1/0_tunis/6/30/8/45/0/-1,1,4.75,initialise
6,bizerte/6/30/7/45/+1/0_tunis/6/30/8/45/0/-1,1,4.75,initialise
7,bizerte/6/30/7/45/+1/0_tunis/6/30/8/45/0/-1,1,4.75,initialise
11,bizerte/6/30/7/45/+1/0_tunis/6/30/8/45/0/-1,1,4.75,initialise
12,bizerte/6/30/7/45/+1/0_tunis/6/30/8/45/0/-1,1,4.75,initialise
13,bizerte/6/30/7/45/+1/0_tunis/6/30/8/45/0/-1,1,4.75,initialise
14,bizerte/6/30/7/45/+1/0_tunis/6/30/8/45/0/-1,1,4.75,initialise
15,sousse/6/30/7/30/+1/0_tunis/9/30/10/30/0/-1,1,5.5,initialise
16,sousse/6/30/7/30/+1/0_tunis/9/30/10/30/0/-1,1,5.5,initialise
17,sousse/6/30/7/30/+1/0_tunis/9/30/10/30/0/-1,1,5.5,initialise
18,sousse/6/30/7/30/+1/0_tunis/9/30/10/30/0/-1,1,5.5,initialise
19,sousse/6/30/7/30/+1/0_tunis/9/30/10/30/0/-1,1,5.5,initialise
23,sousse/6/30/7/30/+1/0_tunis/9/30/10/30/0/-1,1,5.5,initialise
24,sousse/6/30/7/30/+1/0_tunis/9/30/10/30/0/-1,1,5.5,initialise
25,sousse/6/30/7/30/+1/0_tunis/9/30/10/30/0/-1,1,5.5,initialise
26,sfax/10/00/11/30/+1/0_tunis/14/15/16/00/0/-2,1,18,initialise
27,sfax/10/00/11/30/+1/0_tunis/14/15/16/00/0/-2,1,18,initialise
28,sfax/10/00/11/30/+1/0_tunis/14/15/16/00/0/-2,1,18,initialise
29,sfax/10/00/11/30/+1/0_tunis/14/15/16/00/0/-2,1,18,initialise
30,sfax/10/00/11/30/+1/0_tunis/14/15/16/00/0/-2,1,18,initialise
31,sfax/10/00/11/30/+1/0_tunis/14/15/16/00/0/-2,1,18,initialise
32,sfax/10/00/11/30/+1/0_tunis/14/15/16/00/0/-2,1,18,initialise
33,sfax/10/00/11/30/+1/0_tunis/14/15/16/00/0/-2,1,18,initialise
34,sfax/10/00/11/30/+1/0_tunis/14/15/16/00/0/-2,1,18,initialise
35,bizerte/6/20/7/20/+3/0_rassjel/7/20/8/20/0/-3,3,7.5,initialise
36,bizerte/6/20/7/20/+3/0_rassjel/7/20/8/20/0/-3,3,7.5,initialise
37,bizerte/6/20/7/20/+3/0_rassjel/7/20/8/20/0/-3,3,7.5,initialise
38,bizerte/6/20/7/20/+3/0_rassjel/7/20/8/20/0/-3,3,7.5,initialise
39,nabeul/6/20/7/20/+1/0_tunis/8/20/9/20/0/-1,1,6.75,initialise
40,nabeul/6/20/7/20/+1/0_tunis/8/20/9/20/0/-1,1,6.75,initialise
41,nabeul/6/20/7/20/+1/0_tunis/8/20/9/20/0/-1,1,6.75,initialise
42,nabeul/6/20/7/20/+1/0_tunis/8/20/9/20/0/-1,1,6.75,initialise
43,tunis/6/15/7/15/+2/0_bizerte/7/45/8/30/0/-2,2,9.5,initialise
44,tunis/6/15/7/15/+2/0_bizerte/7/45/8/30/0/-2,2,9.5,initialise
45,tunis/6/15/7/15/+2/0_bizerte/7/45/8/30/0/-2,2,9.5,initialise
46,tunis/6/15/7/15/+2/0_bizerte/7/45/8/30/0/-2,2,9.5,initialise
47,bizerte/6/45/7/30/+1/0_elalia/7/45/8/30/0/-1,1,1.5,initialise
48,bizerte/6/45/7/30/+1/0_elalia/7/45/8/30/0/-1,1,1.5,initialise
49,bizerte/6/45/7/30/+1/0_elalia/7/45/8/30/0/-1,1,1.5,initialise
50,bizerte/6/45/7/30/+1/0_elalia/7/45/8/30/0/-1,1,1.5,initialise
51,tunis/6/30/7/30/+1/0_sousse/9/30/10/30/0/-1,1,5.5,initialise
52,tunis/6/30/7/30/+1/0_sousse/9/30/10/30/0/-1,1,5.5,initialise
53,tunis/6/30/7/30/+1/0_sousse/9/30/10/30/0/-1,1,5.5,initialise
54,tunis/6/30/7/30/+1/0_sousse/9/30/10/30/0/-1,1,5.5,initialise
55,tunis/6/20/7/20/+1/0_nabeul/8/20/9/20/0/-1,1,6.75,initialise
56,tunis/6/20/7/20/+1/0_nabeul/8/20/9/20/0/-1,1,6.75,initialise
57,tunis/6/20/7/20/+1/0_nabeul/8/20/9/20/0/-1,1,6.75,initialise
58,tunis/6/20/7/20/+1/0_nabeul/8/20/9/20/0/-1,1,6.75,initialise
59,tunis/09/45/10/30/+1/0_sfax/15/00/16/00/0/-1,1,18,initialise
60,tunis/09/45/10/30/+1/0_sfax/15/00/16/00/0/-1,1,18,initialise
```



Platform java 1.8.0



IntelliJ IDEA 2016.1.1

Conclusions and perspectives

Tests

- Structure of a line in request offer file :

```
id,requestoffer,itineraary{posi[earliestdate/latestdate/pickup/delivery]},driver
{iddriver/ drivinglicence},vehicle{idvehicle/typevehicle/capacityofvehicle},
initialseatsavailable,actualseatsavailable,status
```

```

requêtes_offres - Bloc-notes
Fichier Edition Format Affichage ?
idrequeteoffre,itineraire{posi[dateauplustôt/dateauplustard/pickup/delivery]},chauffeur{idchauffeur/typepermis},vehicule{idvehicule/typevehicule/capacitevehicule},placeslibresinitiales,placeslibresactuelles,etat
0,bizerte/6/45/7/15/+1/0_tunis/6/45/8/15/0/-1,0/d,0/rouge/8,8,8,initialise
1,bizerte/6/45/7/30/+1/0_rassjel/7/00/8/15/0/-3,1/d,1/bleu/8,8,8,initialise
2,tunis/6/45/7/15/+1/0_tunis/9/30/10/30/0/-1,3/d,3/rouge/8,8,8,initialise
3,sfax/9/45/10/15/+1/0_tunis/14/15/15/10/0/-1,5/d,5/rouge/8,8,8,initialise
8,nabeul/6/45/7/15/+1/0_tunis/9/15/10/00/0/-1,4/d,4/rouge/8,8,8,initialise
9,tunis/6/45/7/15/+1/0_bizerte/6/45/8/15/0/-1,6/d,2/rouge/8,8,8,initialise
10,bizerte/6/30/7/30/+1/0_elalja/6/45/7/45/0/rassjel/7/00/8/00/0/-1,2/d,6/bleu/8,8,8,initialise
15,tunis/6/45/7/15/+1/0_sousse/9/30/10/30/0/-1,8/d,7/rouge/8,8,8,initialise
21,tunis/6/45/7/15/+1/0_nabeul/9/15/10/00/0/-1,9/d,8/rouge/8,8,8,initialise
22,tunis/9/45/10/15/+1/0_sfax/14/15/15/10/0/-1,10/d,9/rouge/8,8,8,initialise

```

- Structure of a line in organization file:

```
idjourney, idoffer, itinerary{posi[earlyestdate/latestdate/pickup/delivery]}iddemand,  
listDemandPositions,status,occupiedseats,availableseats,driver{iddriver/licencekind},vehicl  
e{idvehicle/typevehicle/capacityofvehicle}
```

```

OrganizationReturned0 - Bloc-notes
Fichier Edition Format Affichage ?
idcourse, idoffre, itineraire[posit[dateauplustot/dateauplustard/pickup/delivery]]iddemandes, listDemandesPositions, etat, placesoccupees, placeslibres, chauffeur
0,0,bizerte/6/45/7/15/+1/0_bizerte/6/30/7/45/+1/0_bizerte/6/30/7/45/+1/0_bizerte/6/30/7/45/+1/0_bizerte/6/30/7/45/+1/0_bizerte/6/30/7/45/+1/0_bizerte/6/30
1,1,bizerte/6/45/7/20/+1/0_bizerte/6/45/7/20/+3/0_bizerte/6/45/7/20/+3/0_rassjel/7/20/8/15/0/-3_rassjel/7/20/8/15/0/-3_rassjel/7/15/8/15/0/-1,35*36,35|
2,2,sousse/6/45/7/15/+1/0_sousse/6/30/7/30/+1/0_sousse/6/30/7/30/+1/0_sousse/6/30/7/30/+1/0_sousse/6/30/7/30/+1/0_sousse/6/30/7/30/+1/0_sousse/6/30/7/30/+
3,3,sfax/9/45/10/15/+1/0_sfax/10/0/11/30/+1/0_sfax/10/0/11/30/+1/0_sfax/10/0/11/30/+1/0_sfax/10/0/11/30/+1/0_sfax/10/0/11/30/+1/0_sfax/10/0/11/30/+1/0_sfa
4,8,nabeul/6/45/7/15/+1/0_nabeul/6/20/7/20/+1/0_nabeul/6/20/7/20/+1/0_nabeul/6/20/7/20/+1/0_nabeul/6/20/7/20/+1/0_tunis/8/20/9/20/0/-1_tunis/8/20/9/20/0/-1
5,9,tunis/6/45/7/15/+1/0_tunis/6/45/7/15/+2/0_tunis/6/45/7/15/+2/0_tunis/6/45/7/15/+2/0_bizerte/7/45/8/15/0/-2_bizerte/7/45/8/15/0/-2
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8,21,tunis/6/45/7/15/+1/0_tunis/6/20/7/20/+1/0_tunis/6/20/7/20/+1/0_tunis/6/20/7/20/+1/0_tunis/6/20/7/20/+1/0_nabeul/8/20/9/20/0/-1_nabeul/8/20/9/20/0/-1
9,22,tunis/9/45/10/15/+1/0_tunis/9/45/10/30/+1/0_tunis/9/45/10/30/+1/0_sfax/15/0/15/10/0/-1_sfax/15/0/15/10/0/-1_sfax/14/15/15/10/0/-1,39*60,39|tunis/9/45

```

Tests

Functional tests without incrementality



première itération						
config test	$\overline{R_{o,c}}$	$\overline{R_{o,pc}}$	$\overline{R_{o,v}}$	$\overline{R_{d,c}}$	$\overline{R_{d,v}}$	temps d'exécution (ms)
<i>config1</i> : <i>paramOffre</i> = 10/10/0/0/0, <i>paramDemande</i> = 50/50/0/0	5	4	1	47	3	23
<i>config2</i> : <i>paramOffre</i> = 20/20/0/0/0, <i>paramDemande</i> = 100/100/0/0	12	4	4	92	8	43
<i>config3</i> : <i>paramOffre</i> = 100/100/0/0/0, <i>paramDemande</i> = 500/500/0/0	49	10	31	364	136	379
<i>config4</i> : <i>paramOffre</i> = 500/500/0/0/0, <i>paramDemande</i> = 2500/2500/0/0	268	48	184	2028	472	2892
<i>config5</i> : <i>paramOffre</i> = 1000/1000/0/0/0, <i>paramDemande</i> = 5000/5000/0/0	537	94	369	4088	912	7361
<i>config6</i> : <i>paramOffre</i> = 2000/2000/0/0/0, <i>paramDemande</i> = 10000/10000/0/0	1075	186	739	8208	1792	14747

The number of request offer represents 1/5th of request demand

tests with increment

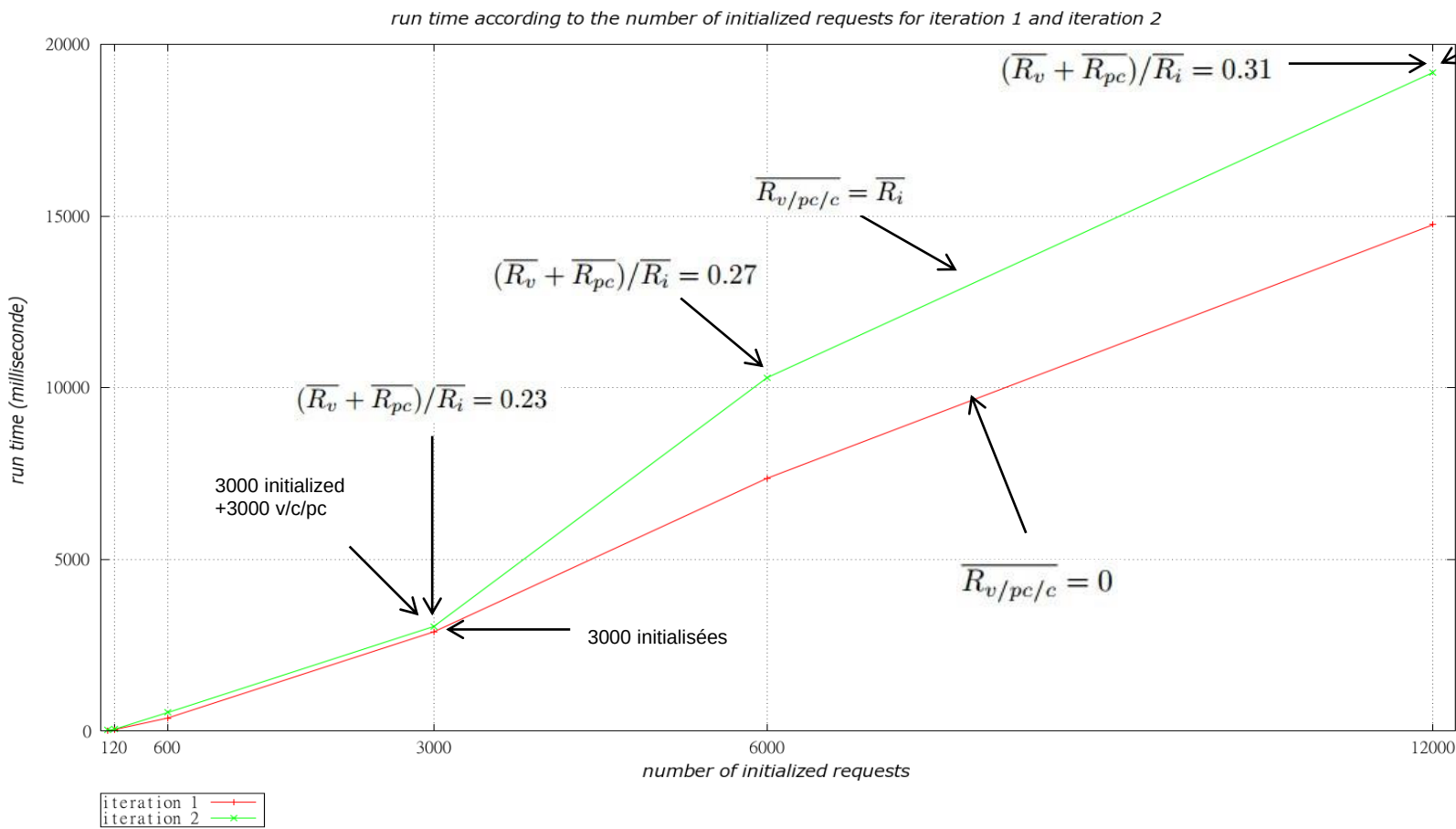


deuxième itération						
config test	$\overline{R_{o,c}}$	$\overline{R_{o,pc}}$	$\overline{R_{o,v}}$	$\overline{R_{d,c}}$	$\overline{R_{d,v}}$	temps d'exécution (ms)
<i>config1</i> : <i>paramOffre</i> = 20/10/1/5/4, <i>paramDemande</i> = 100/50/3/47	10	8	2	93	7	27
<i>config2</i> : <i>paramOffre</i> = 40/20/4/12/4, <i>paramDemande</i> = 200/100/8/92	24	10	6	187	13	50
<i>config3</i> : <i>paramOffre</i> = 200/100/31/49/10, <i>paramDemande</i> = 1000/500/136/364	120	26	54	882	118	538
<i>config4</i> : <i>paramOffre</i> = 1000/500/184/298/48, <i>paramDemande</i> = 5000/2500/472/2028	593	98	309	4416	584	3048
<i>config5</i> : <i>paramOffre</i> = 2000/1000/369/537/94, <i>paramDemande</i> = 10000/5000/912/4088	1075	186	739	8208	1792	10290
<i>config6</i> : <i>paramOffre</i> = 4000/2000/739/1075/186, <i>paramDemande</i> = 20000/10000/1792/8208	1613	278	2109	12328	7672	19183

Legend

config1: *paramOffre* = $\overline{R_o}/\overline{R_{o,i}}/\overline{R_{o,v}}/\overline{R_{o,c}}/\overline{R_{o,pc}}$; *paramDemande* = $\overline{R_d}/\overline{R_{d,i}}/\overline{R_{d,v}}/\overline{R_{d,c}}$

Result analysis



24000 request
4000 louage
40% of number of
louages in 2014
Execution time 19s

✚ The linear evolution of execution time confirm the theoretical complexity of a greedy algorithm of order $O(n)$.

Conclusions

- ✚ Presentation of how louage works in Tunisia.
- ✚ Formalization of louage problem with multi-constraints and multi objectives.
- ✚ Implementation of a greedy pairing algorithm which run in an incremental manner and evaluation (functional and performance).
- ✚ Functionnal validation and benchmark for a centralized deployment in Tunisia

Perspectives

- ✚ Proof of prototype integrating GILA into a management of louage's system.



Thanks for your attention



Publications

- ✚ **Mohamed Hassine**, Philippe Canalda and Idriss Hassine, *Première Résolution Combinatoire Gloutonne du Problème du Louage Incrémental*, In ROADEF'17, Date : 22-24 février 2017, Conference Location : Metz, France
- ✚ **Mohamed Hassine**, Philippe Canalda and Idriss Hassine Tunisian louage's problem : formalization and incremental combinatorial solution, Rairo : Operation Research, pp=19 (papier soumis)
- ✚ **Mohamed Hassine**, Philippe Canalda and Idriss Hassine Tunisian louage's problem : formalization and incremental combinatorial solution, Rapport de recherche FEMTO-ST, pp=14
- ✚ **Mohamed Hassine**, Philippe Canalda and Idriss Hassine Le louage tunisien : une alternative sociétale opérationnelle aux VTC et à la ubérisation à tout-va, FuturMob'17, Montbéliard